

ANNUAL PROGRESS REPORT

Updated through 31 August 2026

Implementing a Multifaceted, Comprehensive, Appropriate Care Package for People with Advanced and Terminal Illnesses Needing ICU Support and Evaluating Its Benefits for Quality of Dying, Family Satisfaction, Care Cost and Resource Utilisation

Report field	Details
Principal Investigator	Dr Naveen Salins
Project/Intervention	MATHRU / Multifaceted Comprehensive Appropriate Care Package (MCACP)
Project start	15 September 2025
Current reporting period	15 September 2025 to 31 August 2026
Overall sanctioned duration	60 months
Current implementation phase	Objective 2: two-ICU pilot (Months 6–18)

Executive progress summary

Overall position on 31 August 2026

The project has completed its evidence synthesis and initial co-development foundation and is actively implementing the two-site pilot. Objective 1 is substantially achieved but not yet formally closed because the protocol-specified two-step Delphi, the final TIDieR description, and measurable usability testing remain outstanding. Objective 2 is in progress within its sanctioned Months 6–18 window. Objectives 3–5 are not yet due.

Objective	Sanctioned period	Status at 31 Aug 2026	Evidence / next requirement
1. Co-develop MCACP	Months 0–6	Substantially achieved; formal closure pending	Six realist reviews; 466 studies; eight stakeholder/expert group discussions; draft package. Two-step Delphi, final TIDieR and formal usability indicators pending.
2. Pilot in two ICUs	Months 6–18	Ongoing	122 quantitative participants (KMC 79; SJMC 43); 66 interviews in consolidated tracker; implementation learning and adaptation ongoing.
3. Expand to four ICUs with palliative-care access	Months 12–36	Not yet due; preparatory work underway	Network, training resources, readiness tools and data systems under preparation.
4. Extend to four ICUs without palliative-care access	Months 24–48	Not yet due	Adaptable pathways and capacity-building content under development.

Objective	Sanctioned period	Status at 31 Aug 2026	Evidence / next requirement
5. Technical Resource Hub	Months 24–60	Not yet due; enabling work initiated	Project website, EPIC curriculum, legal resources and training content in development.

Headline achievements

- Six realist reviews were completed across prognostication, communication, shared decision-making, physician consensus, care delivery and care planning, yielding 466 included studies and an initial realist programme theory.
- Eight multidisciplinary stakeholder/expert group discussions were completed: two with doctors, two with nurses, and one each with allied health professionals, family caregivers, technical site experts and hospital administrators.
- Quantitative baseline recruitment reached 122 participants: 79 at Kasturba Medical College (KMC), Manipal, and 43 at St John’s Medical College Hospital (SJMCH), Bengaluru.
- The consolidated qualitative tracker records 66 in-depth interviews and 62 completed transcriptions; coding and thematic development are ongoing in NVivo 15.
- Baseline evidence demonstrates marked inter-site differences in palliative-care integration, communication and documentation, while family satisfaction is moderate at both sites, and healthcare expenditure is highly variable.
- The MCACP content has been refined around early identification, multidisciplinary prognostication, coordinated family communication, shared decision-making, anticipatory care planning, symptom management, documentation, after-death care and implementation support.

Response to expert-review requirements

Expert review requirement	Position in this updated report
Demonstrate the Objective 1 deliverable	The initial MCACP package and its components are explicitly described; Objective 1 is not labelled fully closed.

Expert review requirement	Position in this updated report
Document two-step Delphi	Not completed by 31 August 2026; retained as a formal closure action.
Document the NGT consensus process	Eight expert/stakeholder group discussions and consolidated findings are reported as NGT-informed consensus input. The report does not overstate preliminary categories as the final consensus.
Provide final TIDieR documentation	Final TIDieR documentation remains pending and is assigned to the Objective 1 closure plan.
Share usability and acceptability indicators	Frontline recommendations are reported. Formal measurable usability/acceptability testing remains pending.
Separate adaptations from protocol deviations	Pilot modifications are classified as planned adaptations and will be maintained in a version-controlled adaptation log; no sanctioned objective, overall design or final timeline has been changed.

Objective 1: Co-develop the MCACP intervention

Sanctioned period: Months 0–6 (initial deliverable due 14 March 2026)

Linked work packages: WP1 Evidence synthesis and intervention development; WP2 MCACP manual and toolkit; WP3 stakeholder engagement and co-design

Methods

A participatory mixed-methods development process was undertaken, informed by the UK Medical Research Council framework for complex interventions, realist methodology, CFIR, Normalisation Process Theory, human-centred design and structured consensus methods. Activities included six realist reviews; multidisciplinary stakeholder/expert discussions; development of the initial programme theory; and iterative drafting of the seven MCACP components and their associated implementation resources.

- Realist evidence synthesis to identify context–mechanism–outcome configurations.

- Stakeholder engagement with doctors, nurses, allied professionals, administrators, caregivers/lived-experience representatives and technical site experts.
- Development of practical guidance, SOPs, algorithms, checklists, documentation templates, educational content, audit indicators and fidelity measures.
- Iterative refinement based on site workflows and emerging qualitative and quantitative evidence.

Work completed

Initial MCACP component	Status/evidence at 31 Aug 2026
Programme theory	The initial realist programme theory was generated from six completed realist reviews and stakeholder input.
Prognostication	Triggers, serial reassessment, multidisciplinary judgement and time-limited treatment-trial principles drafted.
Physician/team consensus	Pre-family-meeting team alignment, escalation pathways and documentation requirements drafted.
Communication and family meetings	Preparation prompts, communication lead, privacy, repeated meetings, empathy and handover documentation incorporated.
Shared decision-making	Patient capacity, representative identification, benefits/burdens, values and iterative decision-making addressed.
Anticipatory care planning	Care-goal documentation, treatment limitations, transfer/home-care planning and portable documentation incorporated.
Care delivery	Symptom assessment, anticipatory medications, nursing comfort measures, dying-phase care and after-death care included.
Care evaluation	Family satisfaction, QODD, cost/resource use, implementation and fidelity indicators identified for evaluation.

Results

The evidence synthesis and stakeholder work converged on early identification of palliative-care needs, structured yet non-deterministic prognostication, consistent multidisciplinary communication, timely family meetings, patient- and family-centred shared decision-making, explicit documentation of care goals, active comfort care after treatment limitation, and institutionally supported implementation. Participants repeatedly cautioned against using prognostic scores as automatic indicators of futility and emphasised that treatment limitation must never be equated with withdrawal of care.

Completion assessment and reasons for outstanding work

Status: substantially achieved, not formally closed

The initial package has been sufficiently developed to support pilot implementation, but the protocol-defined closure evidence remains incomplete. The two-step Delphi has not yet been conducted. The expert group discussions provide structured consensus input but remain preliminary and should not be presented as a completed Delphi study. Final TIDieR documentation and formal usability testing with measurable indicators are also pending. Delayed staff onboarding and the sequential requirements for ethics, site initiation, transcription, analysis and field testing compressed the available development period.

Objective 1 closure plan

1. Freeze and version the current initial MCACP package, with a contents register covering programme theory, manual/SOPs, algorithms, templates, educational resources and implementation workflows.
2. Complete the protocol-specified two-step Delphi and document panel composition, response rates, rating thresholds, item retention/modification and consensus results.
3. Document the NGT-informed expert process separately from the Delphi and preserve meeting records, transcripts and decision trails.
4. Complete the TIDieR checklist for the versioned intervention, including materials, procedures, providers, mode, setting, dose, tailoring, modifications and fidelity assessment.

5. Conduct formal frontline usability testing and report participant characteristics, completion time, item-level feedback, acceptability/appropriateness/feasibility measures and resulting changes.

Objective 2: Pilot the MCACP in two ICUs

Sanctioned period: Months 6–18

Current status: Ongoing and within the sanctioned pilot window

Sites: KMC Manipal and SJMCH Bengaluru

Methods

The two-site mixed-methods pilot combines prospective quantitative baseline assessment, qualitative interviews, implementation observation, workflow mapping, intervention adaptation and preparation for later multicentre evaluation. Adults with advanced or terminal illness or high mortality risk are screened prospectively. Quantitative measures include patient characteristics, organ support, ICU use, palliative care indicators, communication and documentation, symptoms, FS-ICU, QODD, costs, and outcomes. Qualitative interviews are being analysed in NVivo 15 using inductive coding alongside CFIR, NPT and realist constructs.

Work completed

Domain	KMC	SJMCH	Combined/current position
Quantitative recruitment	79 of planned 100	43 of the planned 100	142 participants
Qualitative interviews	36	30	Consolidated: 66
Transcriptions	Included in the combined tracker	Included in the combined tracker	62 completed
Analysis	Interim descriptive and exploratory inferential analysis completed	Interim descriptive analysis completed	Qualitative coding ongoing in NVivo 15

Quantitative findings: KMC (n = 79)

- Mean age was 59.0 ± 14.5 years; 54.4% were male, and 49.4% lived in rural areas.
- End-stage organ failure was the commonest trajectory (49.4%); 86.8% of those with baseline-support data were mechanically ventilated, and 34.2% received vasopressors.

- Specialist palliative-care consultation was documented for 6/79 (7.6%), prognosis discussion for 8/79 (10.1%), advance directives for 2/79 (2.5%) and family meetings for 7/79 (8.9%).
- Median overall FS-ICU was 66.3 (IQR 53.8–76.0). QODD was completed for 29 patients, with a median of 22.6 (IQR 15.6–27.1).
- Median hospital bill among 73 patients was ₹4,42,733 (IQR ₹3,30,573–₹6,31,635). ICU length of stay correlated strongly with cost (Spearman $\rho = 0.766$, $p < 0.001$); ventilation duration showed a modest correlation ($\rho = 0.295$, $p = 0.014$).
- Documented palliative-care consultation was associated with lower median hospital expenditure ($p = 0.037$), but only six consulted patients had cost data; no significant association with overall outcome was detected ($p = 0.203$). These exploratory results do not establish causality.
- Outcomes were: discharge home 30 (38.0%), home care 1 (1.3%), ICU death 29 (36.7%), DAMA 14 (17.7%) and DOR 5 (6.3%). Patients with DAMA were younger ($p = 0.029$), but cost did not differ significantly ($p = 0.970$).

Quantitative findings: SJMCH (n = 30)

- Mean age was 66.2 ± 14.5 years; 56.7% were male, and 86.7% lived in urban areas.
- Other life-limiting ICU conditions were the commonest trajectory (40.0%); vasopressors were used in 60.0%, mechanical ventilation in 56.7%, NIV/HFNC in 26.7%, and dialysis in 10.0%.
- Palliative-care referral was documented for 70.0%, consultation for 63.3% and prognosis for 100%. Family meetings were documented in 70.0%, goals-of-care documents in 53.3%, treatment limitations in 60.0%, and joint ICU–palliative-care–family meetings in 23.3%. No advance directive was recorded.
- Median overall FS-ICU was 65.2 (IQR 52.0–76.0). Ten QODD assessments were completed, with a median of 41.2 (IQR 33.1–62.1).
- Median total ICU cost was ₹2,59,190 (IQR ₹1,57,151–₹4,10,688). Out-of-pocket payment predominated (66.7%).
- Among 27 patients with outcome data, 11 (40.7%) were discharged home, and 11 (40.7%) died in ICU; two (7.4%) entered hospice, and one each was referred elsewhere, transferred to home care, or discharged against medical advice.

Cross-site interpretation

The contrasting sites demonstrate why MCACP must be adaptable. KMC shows limited documented integration and communication of palliative care, whereas SJMCH shows substantially higher documentation of referrals, consultations, prognoses, and family meetings. Nevertheless, both sites show gaps in advance directives and joint multidisciplinary meetings, moderate and variable family satisfaction, substantial symptom burden, and wide cost variation. These differences will inform context-specific implementation strategies while preserving core intervention functions.

Qualitative progress and preliminary findings

Stakeholder group	Interviews	Transcribed	Analysis on 31 Aug 2026
ICU clinicians	9	9	Coding ongoing
ICU nurses	15	15	Coding ongoing
Hospital administrators	8	8	Coding ongoing
Palliative-care clinicians	5	5	Pending/early coding
ICU social workers	6	6	Pending/early coding
Family caregivers	17	14	Transcription and coding pending
Public representatives	6	5	Transcription and coding pending
Total	66	62	Final thematic analysis pending

Source: consolidated category-wise tracker supplied for the 31 August 2026 update. Monthly site totals require reconciliation.

- Palliative care should be introduced early, according to need and trajectory, and should include both cancer and non-cancer illnesses.
- Prognostication requires structured guidance, serial reassessment and multidisciplinary judgement; scores should trigger discussion rather than determine futility.
- Teams should align before family meetings, provide consistent information, identify a communication lead and allow repeated discussions as understanding evolves.

- Nurses possess important bedside knowledge but remain peripheral to formal decision-making; participation must be supported through role clarity, training and feasible workflows.
- Goals-of-care decisions must translate into specific bedside actions, including what will continue, what will stop, symptom relief, dignity, family presence and safe transitions.
- Documentation and handover are essential for continuity; agreed goals, treatment limitations and pending decisions must be distinguished clearly.
- Implementation requires practical multidisciplinary training, institutional policy, legal support, adequate staffing, audit and feedback, and adaptation to open, closed and semi-open ICU models.

Pilot adaptations and protocol fidelity

Modifications arising during the pilot are being treated as planned intervention adaptations that refine delivery without changing the sanctioned objectives, study design, or overall 60-month timeline. Examples include simplifying triggers, tailoring workflows to open and closed ICU models, refining family-meeting guides, clarifying nursing roles, strengthening bedside action plans, and improving documentation prompts. These changes should be entered prospectively in a version-controlled adaptation log that records the date, site, component, rationale, evidence source, decision-maker, expected effect and whether the change affects core function or only form.

Why some Objective 2 activities remain incomplete?

The pilot remains within its sanctioned Months 6–18 window. Recruitment has not reached the planned 100 participants at either site; qualitative coding and thematic integration are ongoing; and consolidated feasibility, acceptability, fidelity, health-economic and readiness assessments are not yet due. Delayed staff onboarding, site-specific ethics and initiation, workflow harmonisation, and the sensitive timing of ICU/family recruitment contributed to slower early progress. These are operational timing issues rather than changes to the sanctioned design.

Objective 3: Expand implementation to four ICUs with palliative-care access

Sanctioned period: Months 12–36

Methods

A multicentre mixed-methods implementation study guided by CFIR and NPT will combine formative context assessment, tailored implementation, process evaluation, iterative adaptation, and quasi-experimental pre–post evaluation of the quality of dying, family satisfaction, cost, and resource utilisation.

Work completed and results

Preparatory work includes the development of the multicentre network, early engagement with participating centres, and the preparation of the intervention package, training resources, readiness tools, governance processes, and data systems. The two-site pilot is generating the specifications, context-sensitive adaptations and fidelity measures required for responsible scale-up. No effectiveness result is yet reportable.

Why has the formal implementation not started

Objective 3 begins at Month 12, approximately 15 September 2026. At the 31 August cut-off, it was not yet due. Initiating a multicentre effectiveness evaluation before stabilising the initial package and completing essential Objective 1 closure work would introduce avoidable heterogeneity in interventions.

Objective 4: Extend implementation to four ICUs without palliative-care access

Sanctioned period: Months 24–48

Methods

The approach will mirror Objective 3, with additional readiness assessment, basic palliative care capacity building, context adaptation, referral pathway development, and evaluation in ICUs without established specialist access.

Status

Formal implementation and outcome evaluation were not scheduled during this reporting period. Evidence synthesis, pilot learning and educational development are producing adaptable resources for lower-readiness settings. There is no schedule deviation on 31 August 2026.

Objective 5: Develop a Technical Resource Hub

Sanctioned period: Months 24–60

Methods

The hub is planned as a modular knowledge-translation and implementation-support platform, progressing through content organisation and peer review, platform selection and design, content upload, training, maintenance/handover arrangements, and development of uptake and outcome metrics.

Preparatory work completed

- The project website is active and being updated: <https://mathru.manipal.edu/>
- Implementation Science 101: Palliative Care in Intensive Care Settings is included on the website.
- The EPIC ICU palliative-care curriculum and associated educational resources are under development.
- Legal and operational resources on advance medical directives and withholding/withdrawal of life-sustaining treatment are being developed with relevant experts.
- Training and advocacy sessions have been conducted at SJMCH, and a practical Living Will resource supporting implementation of Supreme Court directives has been published.
- Realist-review outputs, MCACP tools, training materials and implementation resources are being organised for later hub use.

Status

The objective formally begins at Month 24 and depends on stable, peer-reviewed content generated through Objectives 1–4. Current activities are enabling work for the creation of a national hub of knowledge. Utilisation metrics and handover arrangements are not yet due.

Consolidated results and applied value

Clinical and implementation implications

- Need-based palliative-care triggers should operate early and should not be restricted to cancer or the final days of life.
- Core intervention functions must remain consistent, while workflow form should adapt to local ICU organisation and palliative-care availability.
- The package should combine clinical decision support with communication, documentation, symptom care, transitions, family support and institutional governance.
- The marked inter-site contrast provides an analytically useful test of contextual determinants and implementation strategies.
- Cost findings are promising but preliminary; causal claims about palliative care reducing expenditure are not justified at this stage.

Education, law and capacity building

- Development of the EPIC end-of-life and palliative-care ICU course in collaboration with professional partners.
- Implementation-science and qualitative-research capacity building at participating sites.
- Development and dissemination of practical legal resources for advance medical directives, Primary and Secondary Medical Board processes, and withholding/withdrawal of life-sustaining treatment.
- Training sessions for clinical departments, social workers and caregivers, including advance care planning and Living Will implementation.
- Development of the ApTCare-ICU tool and resources for appropriate transitions after ICU care.

Limitations of current findings

- All quantitative analyses are interim and observational, with incomplete recruitment and variable denominators due to missing data.
- The small number of documented palliative-care consultations at KMC limits precision and creates substantial risk of confounding by illness trajectory and referral timing.
- QODD completion remains limited, especially at SJMCH.
- Qualitative categories are preliminary; coding, comparison, triangulation and final theme development are ongoing.

- The qualitative interview totals require reconciliation between the consolidated tracker and monthly site logs.
- Formal Delphi, TIDieR completion and measurable usability testing remain outstanding.

Corrective action and forward plan

Priority action	Timing/milestone	Evidence to retain
Freeze the initial MCACP version	Immediate	Dated version, contents register and change history
Complete the two-step Delphi	Within the Objective 2 pilot period	Protocol, panel list, response rates, item ratings, thresholds and round-wise decisions
Complete formal NGT documentation	Alongside Delphi reporting	Meeting composition, ranked/prioritised outputs and decision trail
Complete TIDieR description	Before Objective 1 is formally closed	Completed checklist linked to intervention version
Conduct structured usability testing	Before the package freeze for multicentre expansion	Frontline-user sample, measures, task completion, feedback and resulting revisions
Maintain adaptation log	Monthly	Site, date, component, rationale, core function/form, approver and version
Complete pilot recruitment and analyses	Within Months 6–18	Site-wise recruitment, missingness, final quantitative/qualitative/process reports
Prepare Objective 3 initiation	From Month 12	Readiness assessments, ethics/regulatory files, training and baseline systems

Governance safeguards

- Use the sanctioned objective dates as the controlling Gantt chart.

- Report post-Month 6 field refinement under Objective 2, while retaining a versioned initial package as the Objective 1 deliverable.
- Review an objective–work package–deliverable tracker monthly, including recruitment, analysis, document versions and data-quality checks.
- Document the cause, impact, corrective action, owner and completion date for every material delay.
- Seek formal ICMR approval before changing an objective, method, deliverable, budget head or final timeline.

Overall conclusion

By 31 August 2026, the MATHRU programme had completed its core evidence synthesis, established an initial co-developed MCACP, initiated the two-site pilot, enrolled 109 quantitative participants and accumulated a large multidisciplinary qualitative dataset. The updated findings demonstrate substantial contextual variation between the two ICUs and identify actionable priorities for early identification, prognostication, communication, shared decision-making, documentation, comfort care and implementation support. Objective 2 remains active within its sanctioned period, and Objectives 3–5 are not yet due. Objective 1 should be considered substantially achieved but not formally closed until the two-step Delphi, final TIDieR description, protocol-consistent consensus documentation and measurable usability testing are completed and linked to a versioned initial MCACP package. The sanctioned objectives, overall study design, 60-month duration and final deliverable dates remain unchanged.

APPLIED VALUE OF THE PROJECT

1. **Education** – ISCCM (Domain-specific PC in ICU Modules – provide module structure)

EOLC & Palliative Care ICU Course (EPIC)

2. **Workshops at SJMC** – qualitative research workshop, Implementation Science 101:

Palliative care in Intensive Care Settings

3. **Legal framework – with Vidhi Centre for Legal Policy**

In the intensive care unit (ICU), EOLC frequently involves complex decisions regarding withholding or withdrawing life-sustaining treatment, symptom management, communication with families, shared decision-making, and anticipatory care planning. In India, these decisions have traditionally been influenced not only by clinical uncertainty and cultural beliefs but also by concerns regarding legal liability, resulting in considerable variation in clinical practice and delayed integration of palliative care. We are working to simplify the process for healthcare workers by developing a Toolkit that provides hospitals with practical guidance on Advance Medical Directives, withholding and withdrawing life support, documentation, substitute decision-making, conflict resolution, and institutional governance.

23 April 2026: A training session was conducted for the Paediatric ICU and Paediatric Haematology-Oncology Department at SJMC to enhance understanding of the legal and institutional processes involved in withholding and withdrawal of life-sustaining treatment.

8 May 2026: As part of the 25th Anniversary of the Department of Forensic Medicine, a session on the Primary Medical Board (PMB) and Secondary Medical Board (SMB) process was conducted at SJMC. Dr Nandini Vallath delivered a lecture on the withholding and withdrawal of life-sustaining medical treatment, in accordance with the Supreme Court of India's guidelines, to raise awareness and support advocacy.

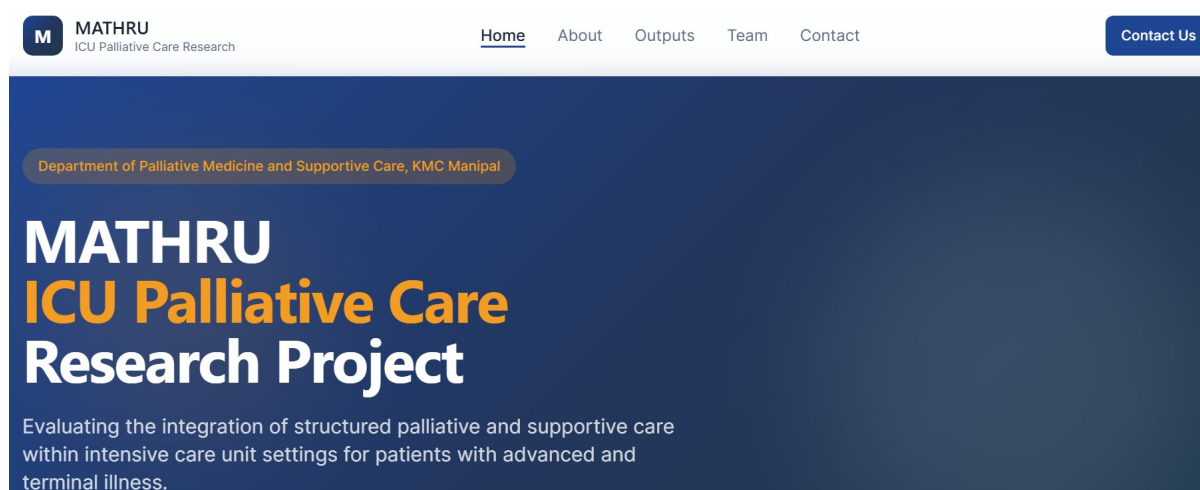
Living Will Document: A Living Will document was developed and published in the Indian Journal of Medical Research to facilitate implementation of the Supreme Court of India's directives on the withholding and withdrawal of life-sustaining medical treatment. The document serves as a practical resource to help individuals and family members record their preferences for future medical care if they are unable to communicate their decisions, thereby supporting advance care planning and informed decision-making.

Karnataka Association of Critical Care: Dr Nandini Vallath conducted a Focus Group Discussion (FGD) on the Primary Medical Board (PMB) and Secondary Medical Board (SMB)

processes, addressing the legal and ethical aspects of withholding and withdrawing life-sustaining medical treatment. The session was held at the CG Hall, St. John's Medical College.

18 June 2026: An educational session was conducted at the St. John's Geriatric Centre for Medical Social Workers (MSWs) and caregivers of patients from St. John's Hospital and the St. John's Geriatric Centre. The session focused on Advance Care Planning, with particular emphasis on the Living Will, its purpose, legal framework, and its role in enabling individuals to document their future healthcare preferences.

4. **ApTCare-ICU (Appropriate Transitions in Care for ICU Patients) tool**, Framework for field notes, two additional forms were developed for patients who recovered for many months after ICU admission (St John's) - ongoing
5. **Project website** - <https://mathru.manipal.edu/>



6. Spin-off projects and academic initiatives

One of the major achievements of the CAR programme has been the development of several postgraduate research projects that have emerged directly from the intervention development and pilot implementation phases. These projects address specific knowledge gaps identified during the realist reviews, stakeholder consultations, and pilot implementation, while contributing additional evidence to strengthen the Multifaceted Comprehensive Appropriate Care Package (MCACP). Collectively, these studies expand the scope of the CAR by generating complementary evidence on governance, education, communication, implementation science, and organisational readiness. The projects also contribute to capacity building by training future clinician-researchers in implementation

science, critical care, and palliative care research. SJMC is working to expand the scope of the project by following up with patients who undergo treatment after transfer from the ICU to a Palliative care setting.

Spin-off Project	Origin within the CAR Programme	Contribution to the CAR
Critical Care Review Board (CCRB) Governance Model Study (PhD)	The physician consensus realist review and stakeholder consultations identified significant variability in multidisciplinary decision-making and a lack of structured governance mechanisms for complex end-of-life decisions. This led to the development of a doctoral project evaluating governance models for multidisciplinary Critical Care Review Boards.	The study will develop evidence-based governance structures, operational policies, accountability frameworks, and decision-making processes that will strengthen the physician consensus component of the MCACP and support wider implementation across Indian ICUs.
ICU Nursing Education in Palliative Care (PhD)	The communication and care delivery reviews highlighted the central role of nurses in symptom assessment, family communication, and continuity of care, while identifying important educational gaps in palliative care competencies.	Findings will inform the development of competency-based educational modules, training resources, and implementation strategies for ICU nurses, strengthening workforce capacity within the MCACP.
Integration of Specialist Palliative Care within Intensive Care (PhD)	The pilot implementation demonstrated limited referrals to specialist palliative care despite a high burden of unmet palliative care needs. This highlighted the	The study will identify effective referral pathways, collaborative care models, and organisational strategies to improve integration of

Spin-off Project	Origin within the CAR Programme	Contribution to the CAR
	need to understand organisational models for integrating specialist palliative care into routine ICU practice. The communication realist review and qualitative interviews	palliative care services, directly informing the care delivery component of the MCACP. Findings will refine communication frameworks, family meeting guides, and shared decision-making strategies within the MCACP
ICU Communication as a Dynamic Field: A Gestalt Analysis of Clinician–Family Interactions (PhD)	identified communication as a dynamic, relational process influenced by contextual, cultural, and emotional factors. This prompted a focused qualitative study exploring clinician–family interactions in ICU settings.	by providing a deeper understanding of the interactional processes that occur during serious-illness conversations.
Assessment of ICU Readiness for Implementing a Multifaceted Palliative Care Intervention: A Mixed-Methods Study (MD Thesis)	During site preparation, considerable variation was observed in organisational readiness, leadership support, staffing, workflows, and institutional culture across participating ICUs. This led to a focused implementation science project evaluating readiness for change.	The study will identify organisational determinants of successful implementation and contribute to the development of readiness assessment tools, implementation checklists, and site-preparation strategies for the future multicentre rollout of the MCACP.
Evaluating the Acceptability, Feasibility, and Fidelity of Early Implementation	Early pilot implementation identified the communication and prognostication bundle as the first components to be integrated into	The study will generate implementation outcome data on acceptability, feasibility, fidelity, and

Spin-off Project	Origin within the CAR Programme	Contribution to the CAR
of the MCACP Communication and Prognostication Bundle in the ICU: A Prospective Observational Study (MD Thesis)	routine ICU practice. Evaluating these components separately provides an opportunity to optimise implementation before wider scale-up.	workflow integration, allowing refinement of intervention delivery and informing the implementation strategy for the multicentre phase of the CAR programme.

Annexure 1 - Intervention Development and drafting the manual

Framework: UK Medical Research Council (MRC) Framework for Complex Interventions

The development of the Multifaceted Comprehensive Appropriate Care Package (MCACP) followed the Medical Research Council (MRC) Framework for the development and evaluation of complex interventions.

Completed activities -

Development phase

- Conducted six realist reviews to identify the mechanisms, contexts, and outcomes influencing appropriate end-of-life care in ICUs.
- Focus group discussions were conducted for each realist review – Doctors, Nurses, Allied Healthcare Professionals, Administrators, Caregivers with Lived Experience, and Technical Site Experts. Stakeholder engagement was coordinated by the CAR research team (CoPIs and research scientists) to contextualise findings from the realist reviews and to ensure that the MCACP reflected the realities of ICU practice in India. Topic guides were developed for each realist review based on the emerging Context–Mechanism–Outcome configurations and were refined following discussions among investigators.
- Purposive sampling was used to recruit stakeholders representing the multidisciplinary ICU team, including intensivists, palliative care physicians, nursing officers, hospital administrators, social workers, psychologists (where available), and family caregivers of critically ill patients. Participants were selected to ensure representation from both open and closed ICU models and from centres with varying levels of palliative care integration (all ten clinical sites).
- Semi-structured interviews and focus group discussions (FGDs) were conducted over MS Teams. Moderators had prior experience conducting qualitative research through the Breathe India programme and received additional orientation regarding realist interviewing techniques and end-of-life care research. An observer (project technical officer) attended each FGD to document group dynamics, non-verbal interactions, and contextual observations.
- Each FGD followed a structured discussion guide exploring current end-of-life care practices, barriers to implementation, multidisciplinary decision-making,

communication processes, organisational culture, legal concerns, documentation practices, and opportunities for improvement. Sessions typically lasted 60–90 minutes and were conducted via MS Teams to facilitate open discussion.

Items	Physicians (number participated)	Nurses (number participated)	Administrators and public health experts (number participated)	Allied health professionals, like physiotherapists, occupational therapists, and psychologists (number participated)	Caregivers/ members of the public with lived experience (number participated)	Key Output for MCACP
Prognostication	13	17	12	10	5	Prognostication methods and contextual factors
Communication	13	17	12	10	5	Communication guide
Shared Decision-Making	13	17	12	10	5	Shared decision-making toolkit
Physician Consensus	13	17	12	10	5	Consensus Care Review Board

						(CCRB) model
Care Planning	13	17	12	10	5	Comprehensive Advance Care Planning template
Care Delivery	13	17	12	10	5	Care delivery pathway, post-ICU care

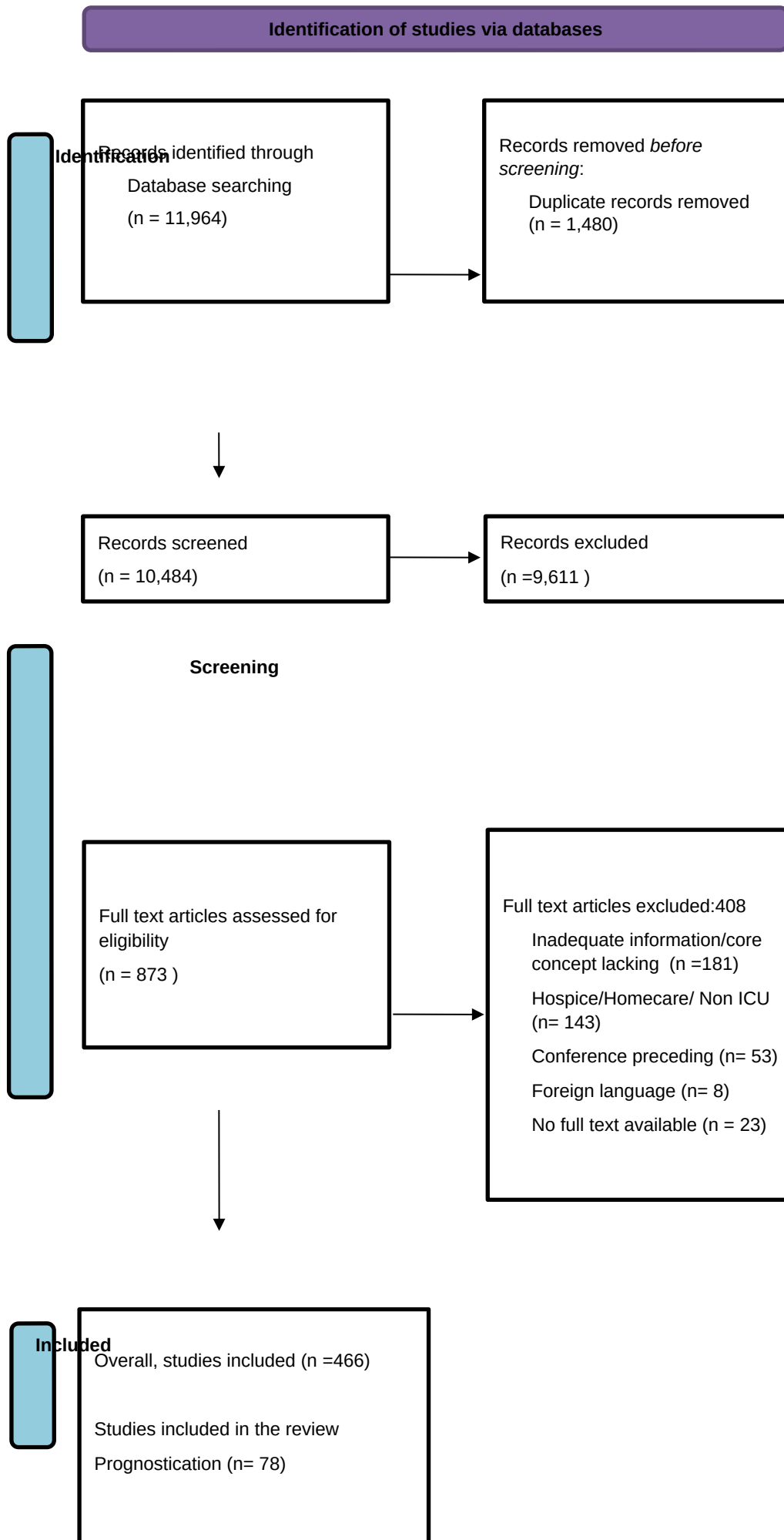
- Following consent, all interviews and FGDs were digitally recorded. Moderators (from the research team) maintained detailed field notes documenting participant interactions, implementation issues, emerging concepts, and contextual observations. Recordings were transcribed verbatim, anonymised, and verified for accuracy by the research team before analysis.
- Qualitative data were managed using NVivo. Coding was undertaken independently by two researchers (from our team) using an inductive–deductive thematic approach informed by realist methodology and the Consolidated Framework for Implementation Research (CFIR). Regular coding meetings were conducted to resolve discrepancies and refine the coding framework. Emerging themes were mapped against the six realist reviews and subsequently synthesised to refine the MCACP intervention.
- Preliminary findings from stakeholder engagement were presented at multidisciplinary investigator meetings (every Friday) that included intensivists, palliative care physicians, and research staff from KMC and SJMC. These meetings enabled iterative refinement of intervention components before pilot implementation. Stakeholder feedback will directly inform the development of prognostication triggers, multidisciplinary review processes, communication pathways, family meeting

templates, documentation formats, symptom management algorithms, and implementation workflows, which will be included in the first draft of the MCACP manual.

- Synthesised evidence into seven preliminary MCACP components:
 - o Prognostication
 - o Physician consensus
 - o Communication
 - o Shared decision-making
 - o Comprehensive anticipatory care planning
 - o Care delivery
 - o Care evaluation
- Developed an initial programme theory describing how each intervention component is expected to improve appropriate care within different ICU contexts.

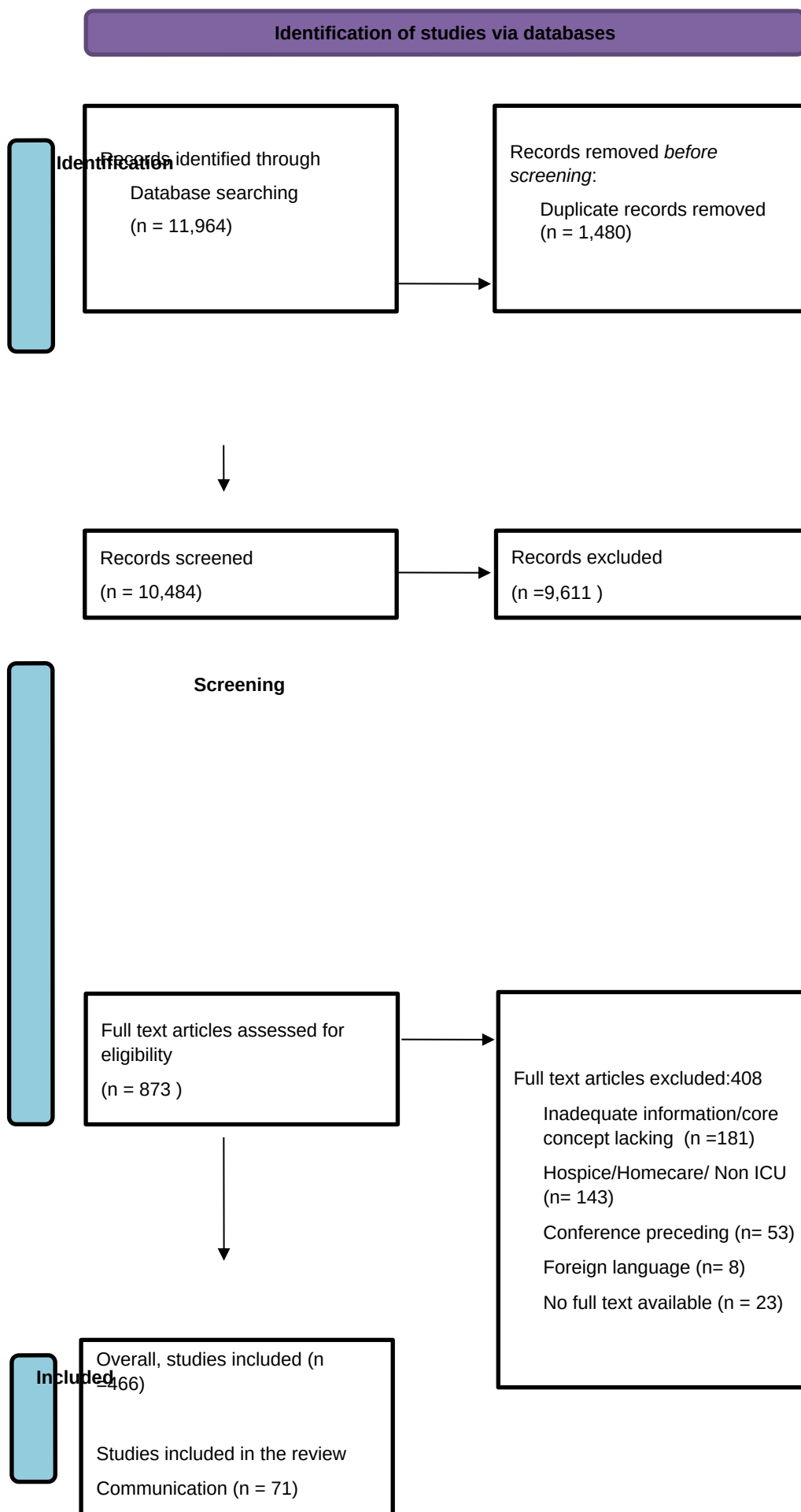
The initial programme theory proposes that the Multifaceted Comprehensive Appropriate Care Package (MCACP) will improve the quality, consistency, and appropriateness of end-of-life care in Indian intensive care units by addressing the clinical, organisational, behavioural, legal, and communication factors that influence decision-making. The intervention recognises that ICU end-of-life care is a complex adaptive process involving multiple stakeholders operating within diverse organisational and cultural contexts. The programme theory assumes that when evidence-based clinical processes are integrated with multidisciplinary collaboration, structured communication, and implementation support, clinicians will be better able to deliver care that aligns with patients' goals, improves family experiences, and reduces non-beneficial interventions.

PROGNOSTICATION



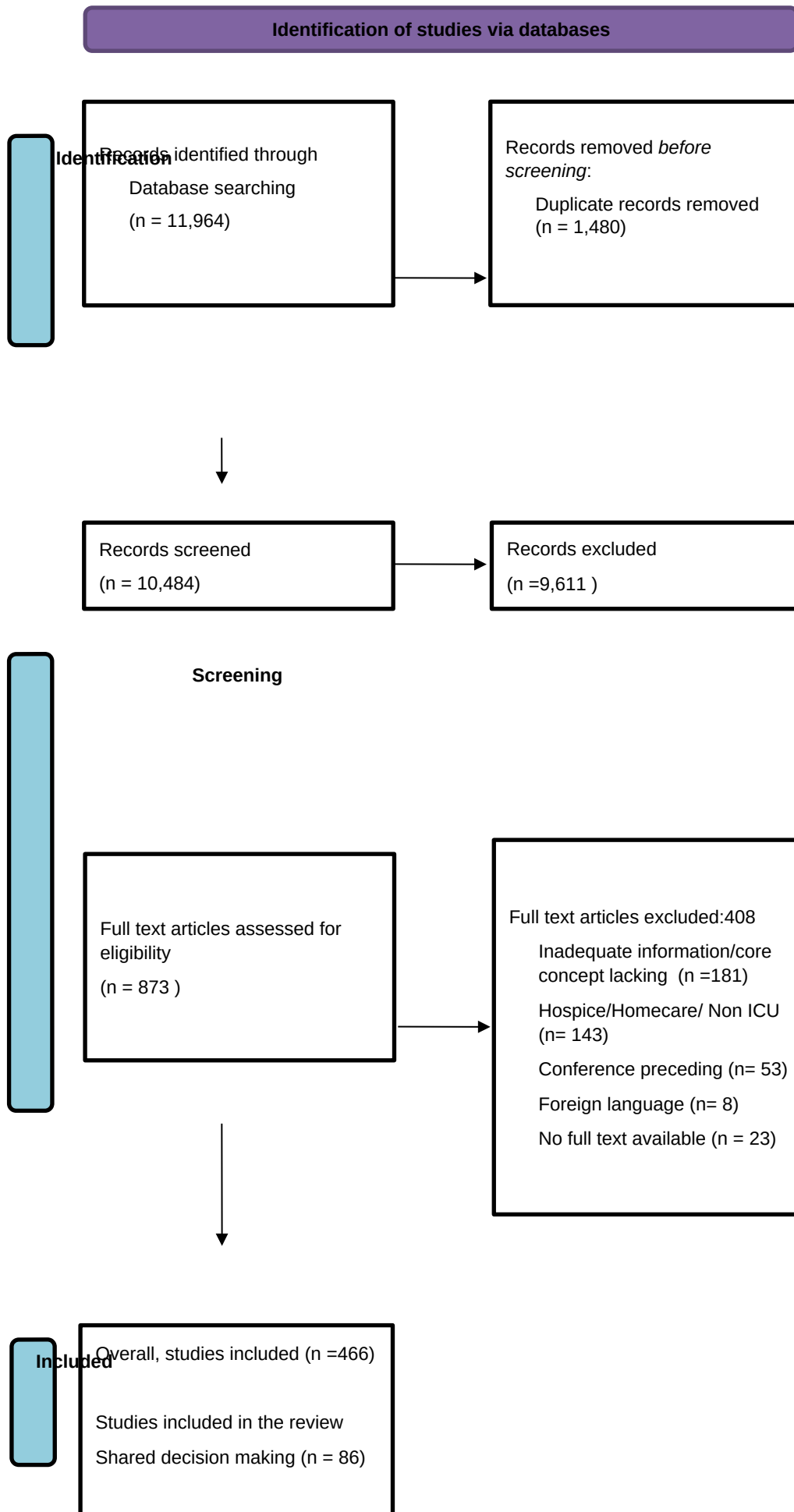
The prognostication realist review identified 78 studies that examined how prognostic assessment influences decision-making in critically ill patients. The evidence suggests that prognostication is a complex and dynamic process that combines clinical expertise, objective prediction tools, patient-specific factors, and interdisciplinary input. The review highlighted that uncertainty in prognosis often delays discussions of goals of care and the integration of palliative care, while structured prognostic approaches may facilitate timely and appropriate treatment decisions.

COMMUNICATION



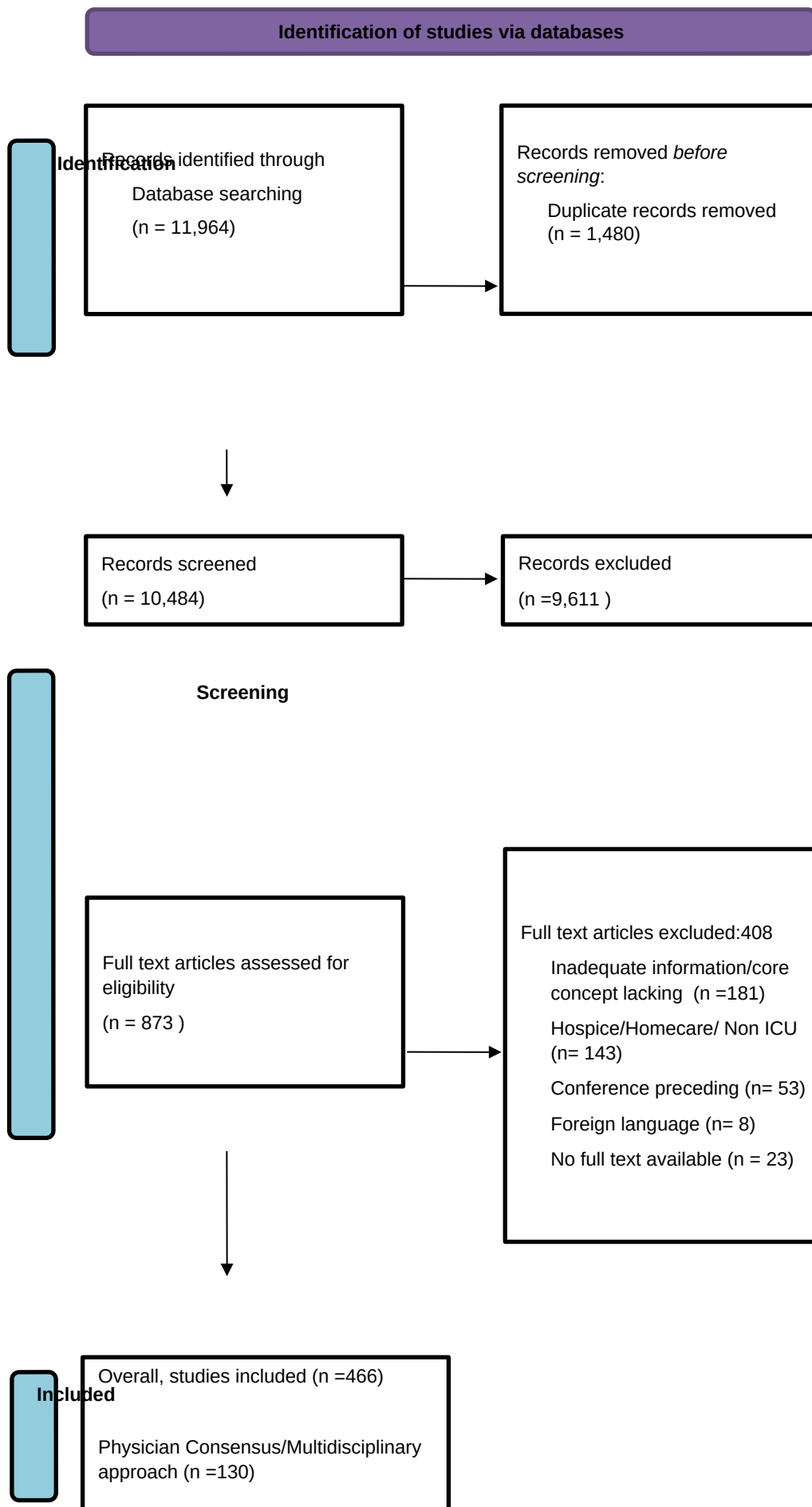
The communication realist review included 71 studies evaluating communication practices between healthcare professionals, patients, and families in the intensive care setting. Preliminary findings indicate that effective communication is central to establishing trust, aligning treatment with patient values, and reducing conflict during critical illness. Interventions such as structured family meetings, communication training, and standardized discussion frameworks were frequently associated with improved satisfaction and better shared understanding of treatment goals.

SHARED DECISION MAKING



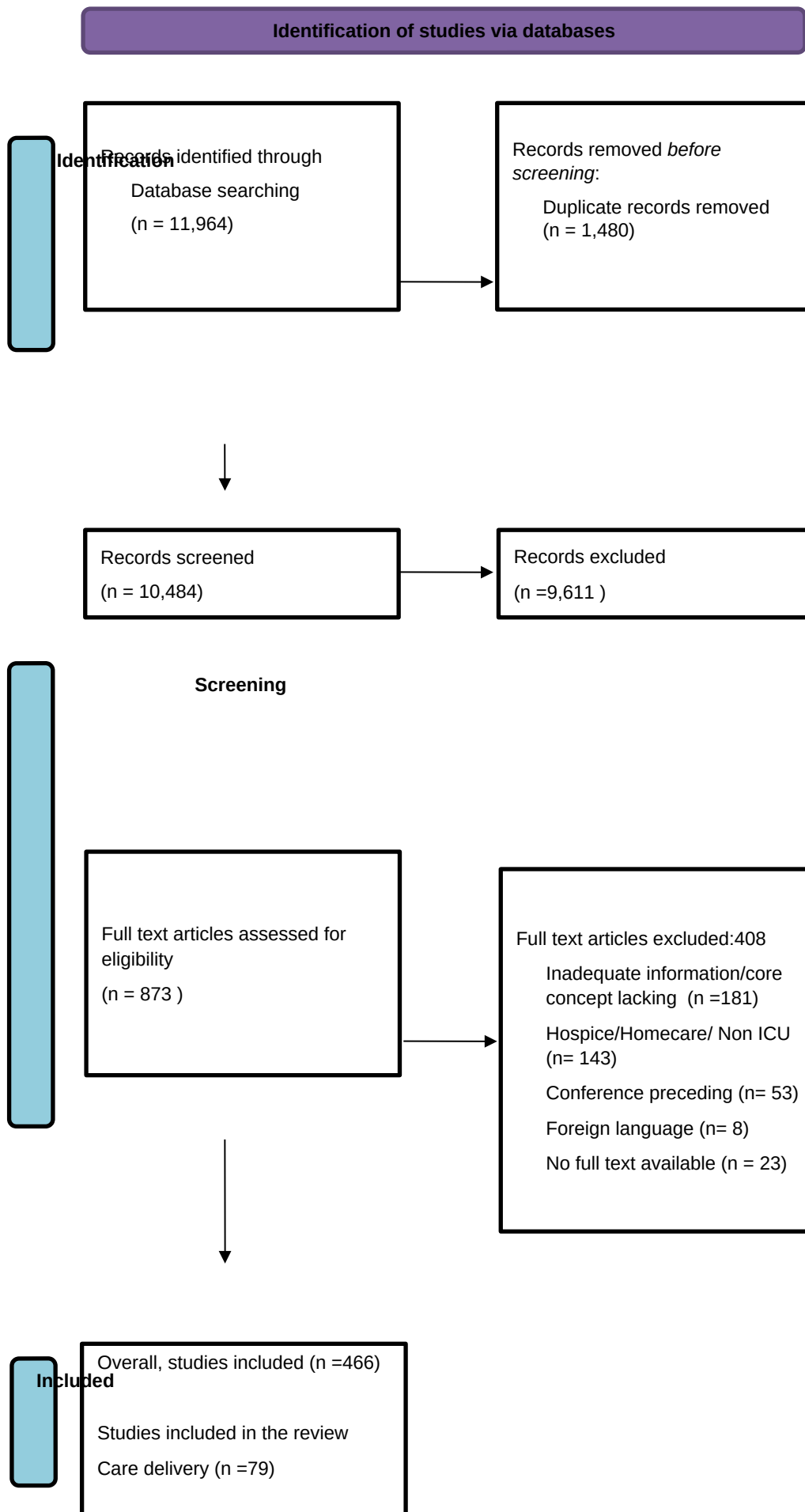
The shared decision-making realist review synthesized evidence from 86 studies examining how patients, families, and healthcare teams participate in treatment decisions. The findings suggest that successful shared decision-making depends on clear communication, accurate prognostic information, family engagement, and clinician willingness to explore patient preferences. Decision-making was found to be most effective when supported by multidisciplinary collaboration and when adequate time and resources were available for meaningful discussions.

PHYSICIAN CONSENSUS/MULTIDISCIPLINARY APPROACH



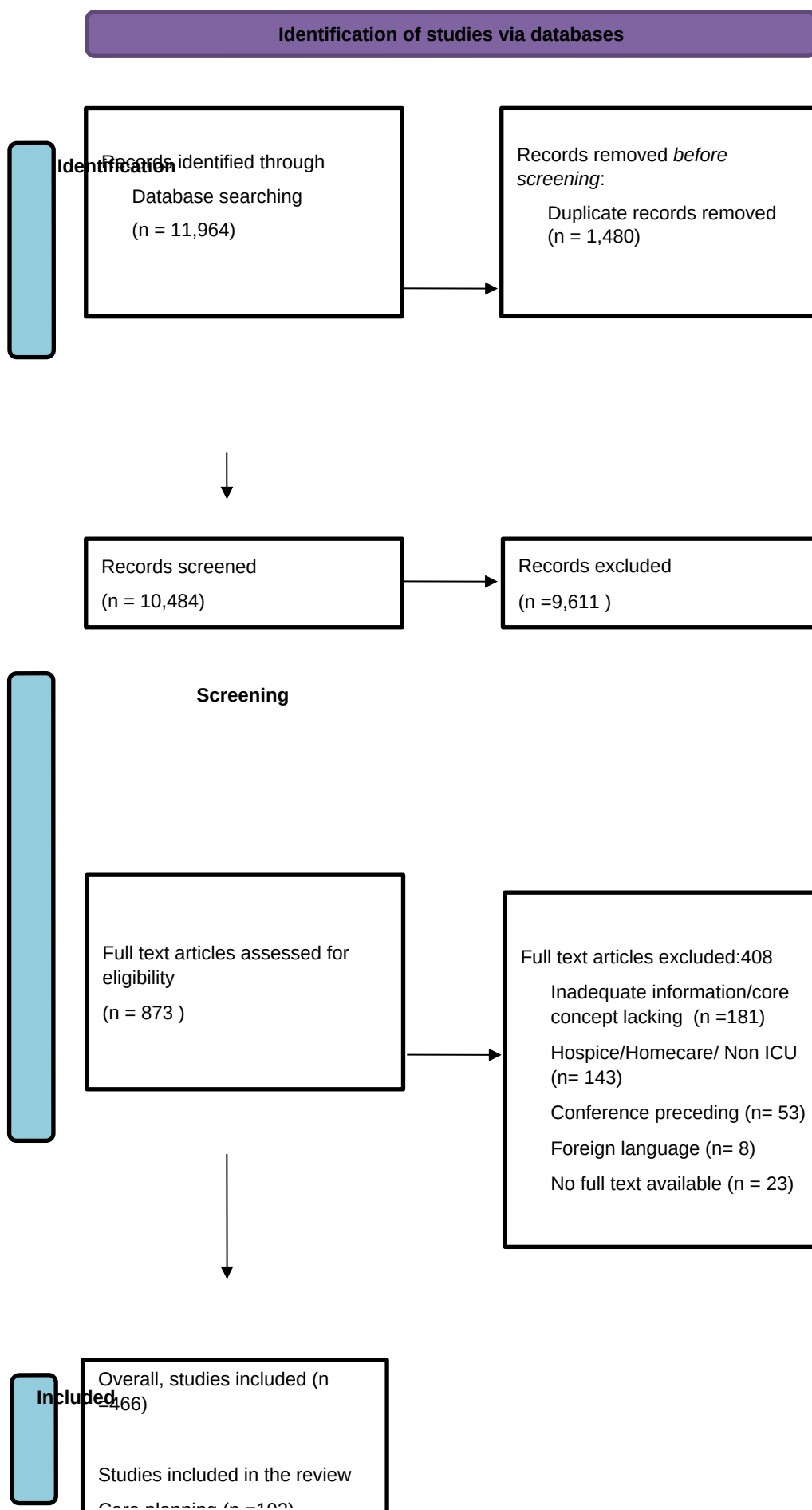
The physician consensus and multidisciplinary approach realist review represented the largest body of evidence, comprising 130 studies. Preliminary findings demonstrate that coordinated multidisciplinary decision-making, including contributions from intensivists, palliative care specialists, nurses, and other allied health professionals, promotes more consistent, patient-centred care. Regular multidisciplinary meetings and consensus-building processes were associated with improved treatment planning, reduced variability in clinical practice, and greater confidence in complex end-of-life decisions.

CARE DELIVERY



The care delivery realist review identified 79 studies evaluating models of integrating palliative care into intensive care practice. The evidence suggests that care delivery is strengthened through early identification of patients with palliative care needs, standardised referral pathways, multidisciplinary collaboration, and organisational support. Integrated models of care were consistently associated with improvements in symptom management, coordination of care, family support, and efficient utilisation of healthcare resources.

CARE PLANNING



The care planning realist review included 102 studies focusing on advance care planning, goals-of-care discussions, and individualised treatment planning. Preliminary findings indicate that proactive care planning facilitates treatment decisions that are better aligned with patients' values and preferences while reducing unnecessary interventions at the end of life. Successful care planning was supported by early conversations, repeated reassessment throughout the illness trajectory, comprehensive documentation, and active involvement of patients, families, and multidisciplinary healthcare teams.

Realist Review	Studies	
	Included (n)	Preliminary Result
Prognostication	78	Seventy-eight studies were identified that addressed prognostication in ICU and palliative care settings, providing evidence on prognostic tools, clinician judgement, and outcome prediction.
Communication	71	Seventy-one studies focused on communication between healthcare professionals, patients, and families, examining approaches to goals-of-care discussions and information sharing.
Shared Decision-Making	86	Eighty-six studies addressed shared decision-making, exploring patient and family involvement in treatment decisions, advance care planning, and collaborative decision processes.
Physician Consensus / Multidisciplinary Approach	130	This was the largest evidence base, with 130 studies examining multidisciplinary collaboration, consensus-building among clinicians, ethics consultation, and interdisciplinary ICU decision-making.
Care Delivery	79	Seventy-nine studies evaluated models of care delivery, including integration of palliative care into ICU

Realist Review	Studies	
	Included (n)	Preliminary Result
		practice, service organisation, and implementation strategies.
Care Planning	102	One hundred and two studies focused on care planning, including advance care planning, goals-of-care documentation, and individualised treatment planning.

Component 1. Prognostication

Context

ICU clinicians often experience uncertainty when identifying patients approaching the end of life, particularly in the presence of multiple comorbidities, organ failure, and unpredictable illness trajectories. This uncertainty frequently delays palliative care referral and goals-of-care discussions.

Mechanism

Providing structured prognostic assessment tools, clinical triggers, and multidisciplinary review increases clinicians' confidence in recognising limited prognosis, facilitates earlier identification of patients requiring appropriate care, and supports timely initiation of advance care planning.

Expected Outcomes

- Earlier recognition of patients with a limited prognosis.
- Timely palliative care referral.
- Earlier goals-of-care discussions.
- Reduction in potentially non-beneficial intensive treatments.

Component 2. Physician Consensus

Context

End-of-life decisions in ICUs are often influenced by differences of opinion among treating

specialists, intensivists, and consulting teams, leading to inconsistent treatment recommendations and delayed decision-making.

Mechanism

Structured multidisciplinary physician review and consensus processes improve shared understanding of prognosis, reduce professional uncertainty, distribute decision-making responsibility, and enhance confidence in treatment limitation decisions.

Expected Outcomes

- Increased multidisciplinary agreement.
- Reduced variation in clinical decisions.
- Timelier treatment limitation decisions.
- Improved consistency of patient-centred care.

Component 3. Communication

Context

Communication regarding prognosis and treatment goals is often delayed, inconsistent, or unstructured, leading to misunderstanding, family conflict, and unrealistic expectations.

Mechanism

Structured communication guides, scheduled family meetings, and communication training improve clinicians' ability to explain prognosis, address uncertainty, explore patient values, and build trusting relationships with families.

Expected Outcomes

- Improved quality of family communication.
- Greater family understanding of prognosis.
- Reduced decisional conflict.
- Increased family satisfaction with ICU care.

Component 4. Shared Decision-Making

Context

Treatment decisions frequently remain clinician-driven because patients are unable to participate, family members have limited understanding of illness, or there is no structured decision-making process.

Mechanism

Decision aids, structured family engagement, and shared decision-making frameworks enable clinicians and families to jointly evaluate treatment options considering patients' preferences, values, and goals of care.

Expected Outcomes

- Increased family participation in decision-making.
- Greater concordance between treatments and patient preferences.
- Reduced moral distress among clinicians and families.
- Improved quality of end-of-life decisions.

Component 5. Comprehensive Anticipatory Care Planning (CACP)**Context**

Advance care planning and documentation of treatment preferences are uncommon in Indian ICUs, resulting in repeated crisis-based decision-making and inconsistent care.

Mechanism

Standardised anticipatory care planning templates and documentation processes facilitate proactive discussions about goals of care, treatment ceilings, advance directives, and future clinical decisions.

Expected Outcomes

- Increased documentation of care preferences.
- Earlier care planning discussions.
- Improved continuity across treating teams.
- Greater delivery of preference-concordant care.

Component 6. Care Delivery**Context**

Patients approaching the end of life often experience high symptom burden, fragmented palliative care involvement, and variable standards of supportive care.

Mechanism

Standardised symptom management protocols, early integration of specialist palliative care, multidisciplinary care coordination, and psychosocial support improve the quality and consistency of patient-centred care.

Expected Outcomes

- Improved symptom control.
- Earlier palliative care involvement.
- Improved patient comfort and dignity.
- Reduced unnecessary invasive interventions.
- Better support for families during the dying process.

Component 7. Care Evaluation

Context

Most ICUs lack routine systems to evaluate the quality of end-of-life care, limiting opportunities for continuous quality improvement.

Mechanism

Routine measurement of implementation fidelity, quality-of-dying indicators, family satisfaction, communication quality, and resource utilisation enable clinicians and organisations to monitor performance, identify gaps, and continuously refine practice.

Expected Outcomes

- Improved adherence to MCACP.
- Higher quality-of-dying scores.
- Greater family satisfaction.
- Continuous quality improvement.
- Sustainable integration of appropriate care into routine ICU practice.

Feasibility phase (ongoing)

- Pilot implementation initiated in two tertiary ICUs with existing palliative care integration.
- Mixed-methods evaluation undertaken to determine feasibility, acceptability, usability, workflow integration, and implementation fidelity.
- Iterative refinement of intervention components based on stakeholder feedback and implementation experience.

The project is therefore progressing from the Development to the Feasibility phase of the MRC framework.

2. Context Assessment

Framework: Consolidated Framework for Implementation Research (CFIR)

CFIR has been used to understand contextual factors influencing implementation across the two pilot ICUs.

Intervention Characteristics

- Assessed the adaptability of MCACP to both open (KMC) and closed (SJMC) ICU models.
- Evaluated perceived complexity of intervention components.
- Identified opportunities to simplify documentation and workflows.

Outer Setting

- Explored patient and family expectations regarding end-of-life care.
- Identified legal and ethical influences affecting treatment limitation decisions.
- Examined the role of institutional policies and national end-of-life care guidance.

Inner Setting

- Assessed organisational culture regarding palliative care integration.
- Explored leadership engagement and multidisciplinary collaboration.
- Evaluated existing communication practices and ICU workflows.

Characteristics of Individuals

- Interviewed ICU physicians, nurses, palliative care clinicians, and administrators regarding:
 - o knowledge
 - o beliefs
 - o confidence
 - o perceived roles
 - o willingness to implement MCACP

Implementation Process

- Established local implementation teams.
- Initiated workflow testing.
- Conducted implementation workshops.
- Used continuous feedback to refine intervention delivery.

3. Co-design and Pilot Refinement

Framework: Human-Centred Design (HCD) + CFIR

The MCACP is being refined using a human-centred design approach that actively involves frontline users throughout implementation.

Completed activities include:

- Stakeholder interviews across both ICUs.
- Continuous workflow observation.
- Identification of implementation barriers and facilitators.
- Iterative refinement of:
 - o referral triggers
 - o communication pathways
 - o documentation templates
 - o multidisciplinary review processes
 - o symptom management workflows

Pilot findings have demonstrated the need for:

- structured family meeting templates
- earlier prognostication
- standardised documentation
- simplified care planning tools
- improved integration of specialist palliative care.

The intervention continues to undergo iterative refinement based on user experience.

4. Behavioural Implementation

Framework: Normalization Process Theory (NPT)

NPT has been used to understand how MCACP becomes embedded into routine ICU practice.

Coherence

Clinicians generally recognise the importance of improving end-of-life care and understand the purpose of MCACP.

Cognitive Participation

ICU physicians, nurses, palliative care teams, and administrators have participated in implementation workshops and qualitative interviews, demonstrating early engagement with the intervention.

Collective Action

Workflow testing is evaluating how MCACP components integrate into daily ICU practice, including multidisciplinary meetings, prognostication, communication, documentation, and symptom management.

Reflexive Monitoring

Continuous feedback from implementation teams is informing ongoing refinement of the intervention, ensuring that it remains practical, acceptable, and contextually appropriate.

5. Implementation Planning

Framework: Implementation Research Logic Model (IRLM)

Implementation planning has followed the IRLM by explicitly linking determinants, strategies, mechanisms, implementation outcomes, and clinical outcomes.

Determinants

Identified determinants include:

- limited structured prognostication
- infrequent palliative care referral
- inconsistent documentation
- absence of formal family meetings
- organisational workflow constraints
- clinician uncertainty regarding end-of-life decision-making

Implementation Strategies

Strategies currently being tested include:

- multidisciplinary implementation workshops
- workflow redesign
- nurse/clinician education
- structured communication tools
- standardised documentation
- trying to identify/assign implementation champions
- continuous audit and feedback

Mechanisms

These strategies are expected to improve:

- nurse/ clinician knowledge
- confidence

- multidisciplinary collaboration
- shared decision-making
- communication consistency
- organisational readiness

Implementation Outcomes

Current evaluation includes:

- feasibility
- acceptability
- fidelity
- usability
- workflow integration
- stakeholder engagement

Clinical Outcomes

Outcomes being monitored include:

- quality of dying
- family satisfaction
- symptom control
- palliative care utilisation
- treatment limitation practices
- resource utilisation

6. Implementation Strategies

Framework: ERIC (Expert Recommendations for Implementing Change)

Several ERIC implementation strategies have already been incorporated into the pilot implementation.

ERIC Strategy	Current Application
Conduct educational meetings	Implementation workshops were conducted at ICUs
Develop educational materials	Preliminary MCACP protocols and training resources are being developed
Identify and prepare	Local ICU and palliative care implementation leads are being

ERIC Strategy	Current Application
champions	identified
Conduct a local needs assessment	Baseline quantitative and qualitative assessments ongoing
Audit and provide feedback	Ongoing monitoring of recruitment, workflow, and implementation progress
Tailor strategies	Intervention adapted for open and closed ICU models
Facilitation	Research team providing continuous implementation support
Build stakeholder relationships	Active engagement of physicians, nurses, administrators, and families
Capture and share local knowledge	Qualitative interviews informing intervention refinement
Revise professional roles	Strengthening multidisciplinary collaboration and physician consensus pathways

Annexure 2 – Pilot the MCACP Intervention in Two ICUs with Existing Palliative Care Integration

Work Package 2.1: Baseline Evaluation and Contextual Assessment

Methodology

A mixed-methods baseline assessment is being undertaken in two tertiary ICUs with established palliative care services (KMC Manipal and St. John's Medical College, Bengaluru). The objective is to understand existing ICU practices related to appropriate care, identify contextual determinants of implementation, and establish baseline patient, family, clinician, and organisational outcomes. The study follows the UK Medical Research Council framework for complex interventions and is guided by the Consolidated Framework for Implementation Research (CFIR) and Normalisation Process Theory (NPT).

Participant Recruitment

Patients were screened daily by trained research staff in collaboration with ICU clinicians. Eligible participants included adult ICU patients meeting predefined criteria for advanced life-limiting illness or those considered at high risk of mortality based on clinical judgement and ICU admission characteristics. Recruitment was undertaken using consecutive sampling to minimise selection bias.

Data Collection

Dedicated research coordinators employed through the CAR programme conducted daily data collection under the supervision of the site principal investigators.

Clinical information was obtained through:

- electronic medical records
- bedside patient assessment
- ICU observation charts
- nursing documentation
- laboratory records
- medication charts
- family discussions where appropriate

Standardised Case Report Forms (CRFs) were developed specifically for the project and piloted prior to commencement – see Annexure 6. Case Record Proforma.

Variables collected included:

Patient Characteristics

- age
- gender
- diagnosis
- illness trajectory
- comorbidities
- frailty
- reason for ICU admission

ICU Care

- mechanical ventilation
- vasopressor support
- renal replacement therapy
- invasive procedures
- ICU length of stay

Palliative Care Indicators

- specialist palliative care referral
- prognosis discussions
- family meetings
- treatment limitation decisions
- advance directives
- goals-of-care documentation

Symptom Assessment

- pain
- dyspnoea
- agitation
- respiratory secretions
- delirium

Family Outcomes

- family satisfaction
- communication quality
- quality of dying

Daily monitoring continued until ICU discharge or death.

AIM, IAM and FIM calculation

AIM, IAM, and FIM are three separate, validated, four-item implementation measures completed by staff exposed to the intervention.

Response scale

Each item uses a five-point Likert response:

1. Completely disagree
2. Disagree
3. Neither agree nor disagree
4. Agree
5. Completely agree

Domains

Measure	What it assesses	Typical item concepts
AIM	Acceptability	Approval, appeal, liking and welcoming the intervention
IAM	Appropriateness	Fit, relevance, suitability and applicability
FIM	Feasibility	Implementability, possibility, practicality and ease of use

Individual respondent score

For each scale:

$$\text{Individual scale score} = \frac{\text{sum of four item scores}}{\text{number of completed items}}$$

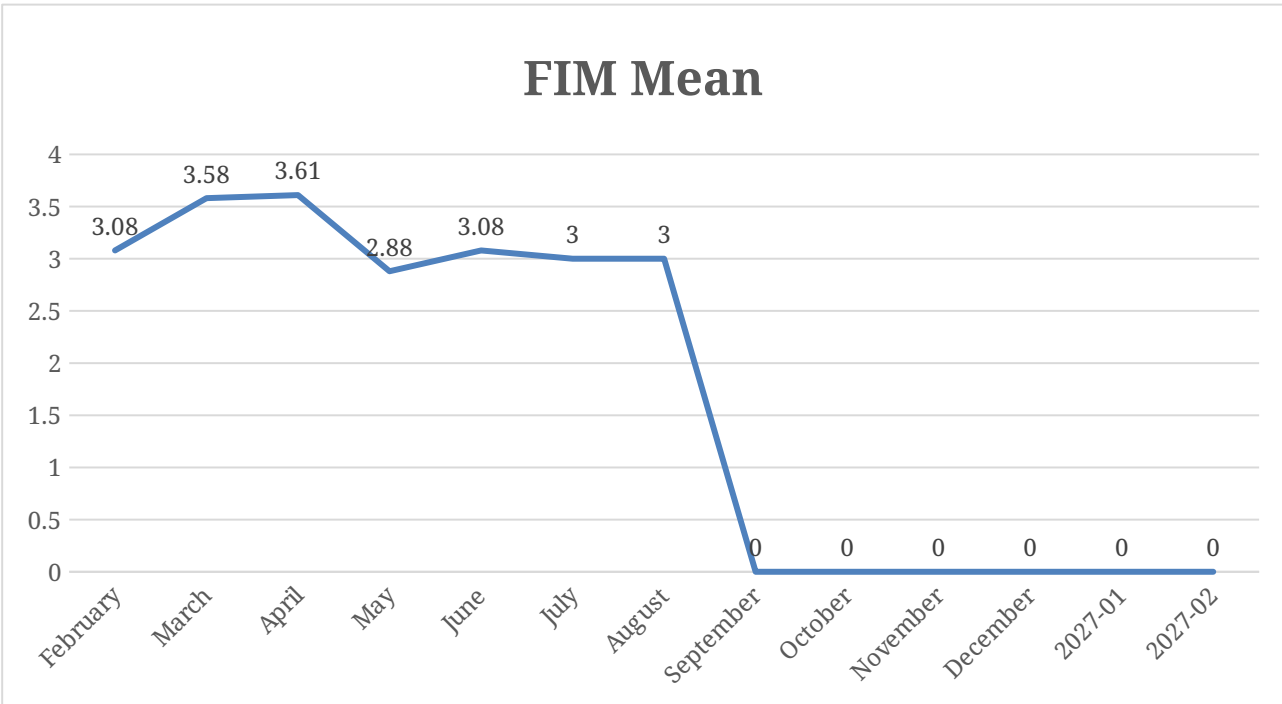
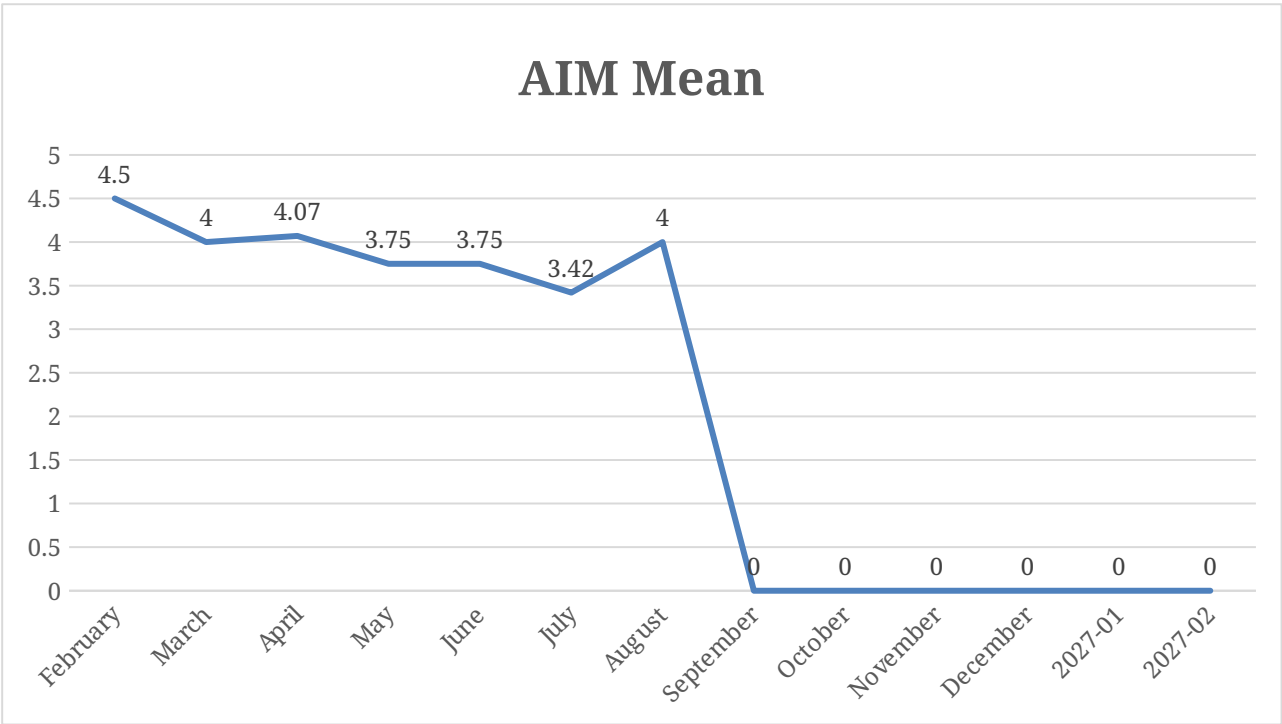
Monthly mean

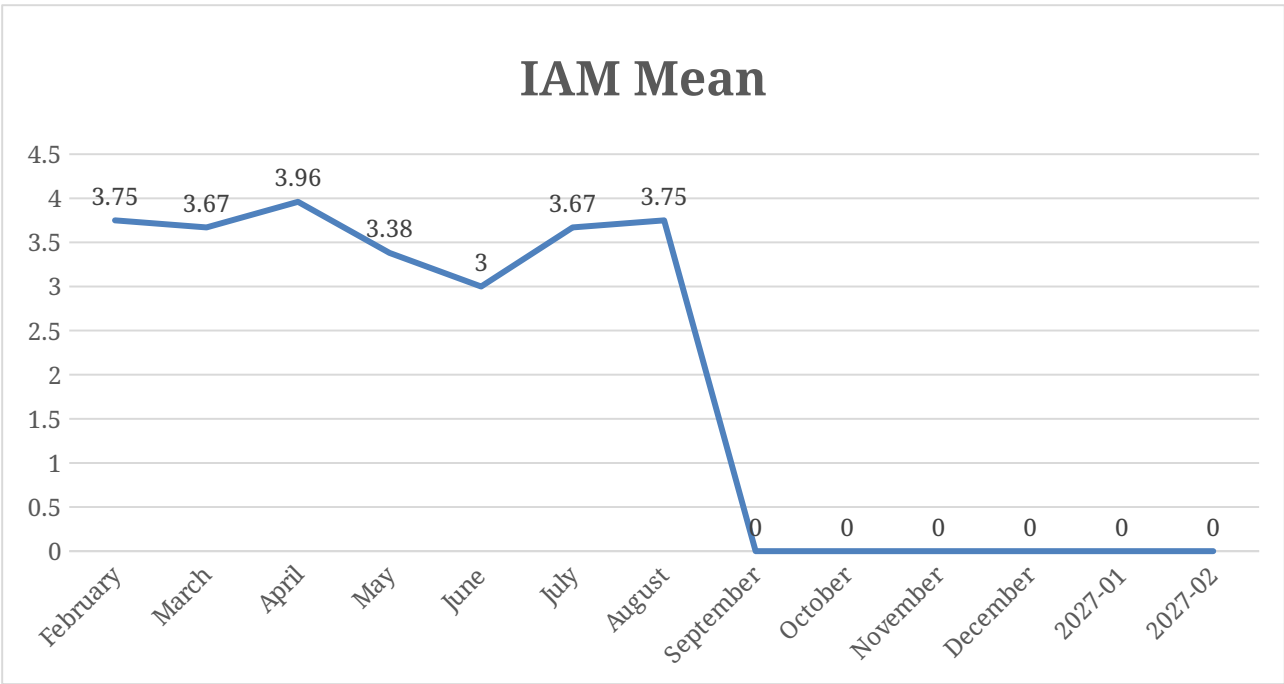
$$\text{Monthly mean} = \frac{\sum \text{individual respondent scale scores during the month}}{\text{number of valid respondents during the month}}$$

Interpretation

Scores range from **1 to 5**:

- **1–1.9**: strong concern;
- **2–2.9**: generally unfavourable;
- **3**: neutral or uncertain;
- **3.1–3.9**: moderately favourable;
- **4–5**: favourable.

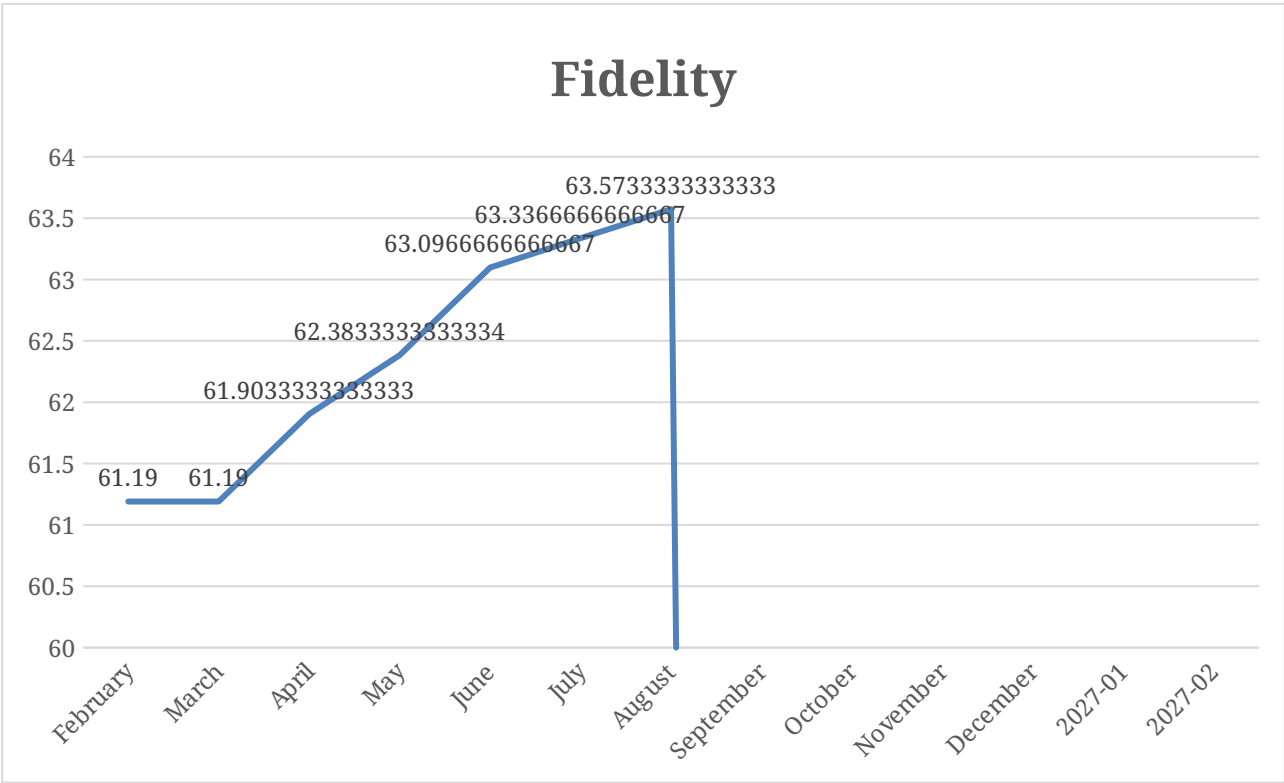




Fidelity calculation

Fidelity indicates the extent to which the core MCACP actions were delivered as intended.

Patient-level fidelity percentage = (applicable required actions completed / total applicable required actions) × 100



Adoption calculation

Patient-level adoption = $\frac{\text{eligible patients for whom MCACP was activated}}{\text{total eligible patients}} \times 100$

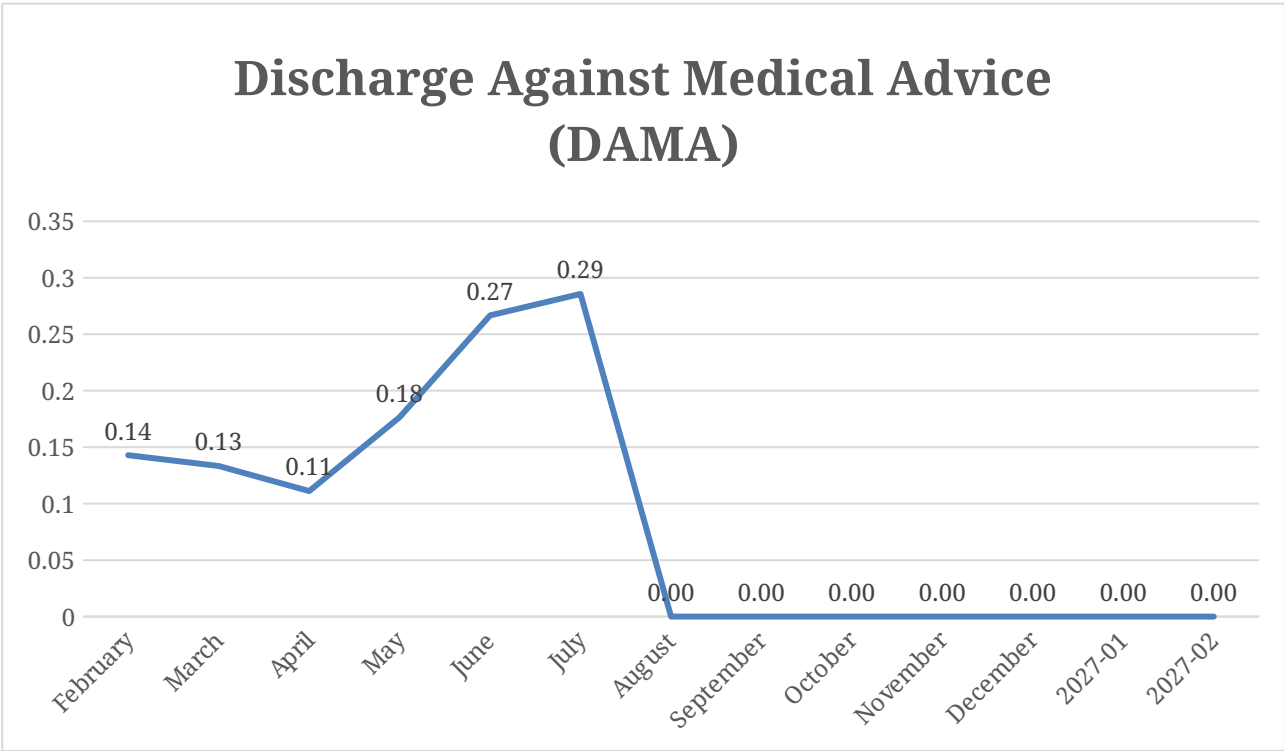
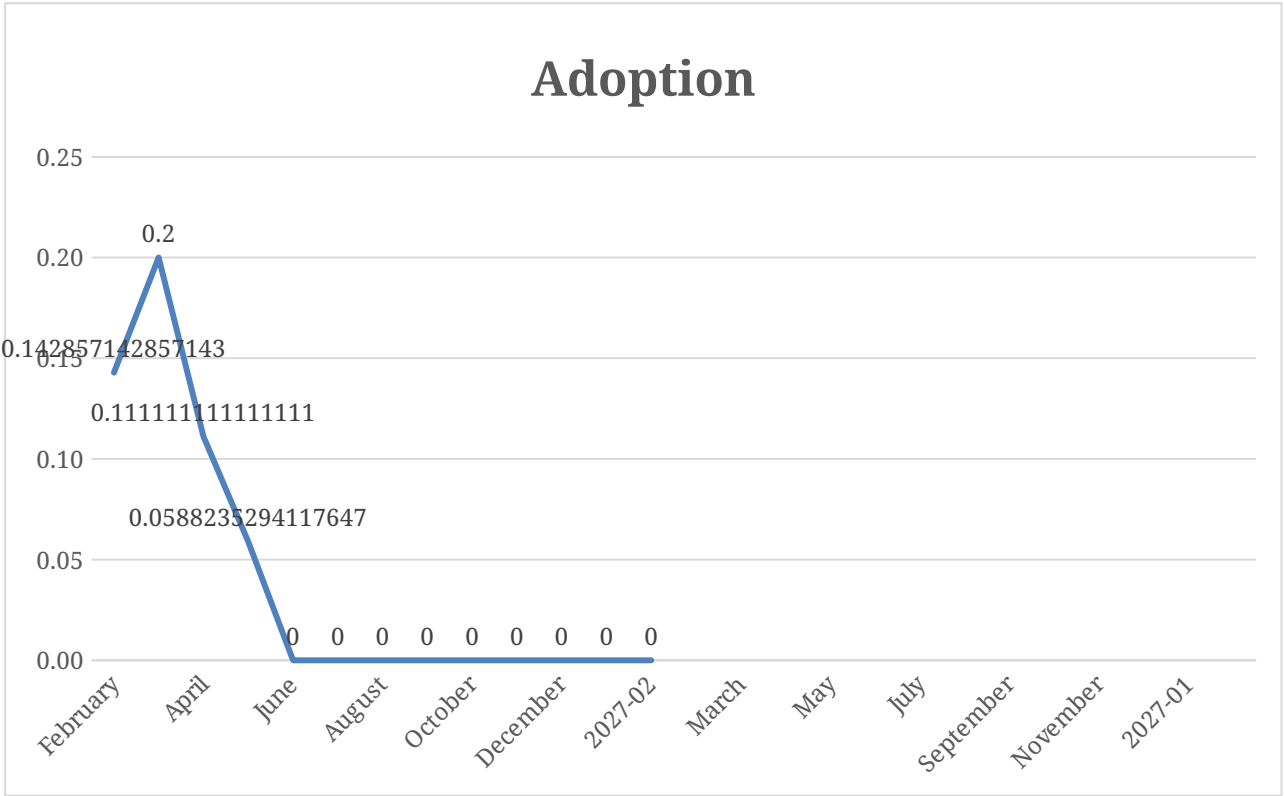


Figure description: Monthly implementation and outcome indicators from February to August 2026. Adoption varied across the initial implementation period, increasing from 0.14 in February to a peak of 0.20 in March, then declining to 0.11 in April and 0.06 in May, with no adoption recorded from June onward. Fidelity improved progressively from 61.19% in February and March to 63.57% in August. The proportion of discharges against medical advice (DAMA) declined from 0.14 in February to 0.11 in April, then increased to 0.18 in May, 0.27 in June, and peaked at 0.29 in July, with reporting through August.

Implementation outcome scores remained generally moderate to high. The Acceptability of Intervention Measure (AIM) mean ranged from 3.42 to 4.50, declining gradually through July before increasing to 4.00 in August. The Feasibility of Intervention Measure (FIM) mean ranged from 2.88 to 3.61 and remained at 3.00 in July and August. The Intervention Appropriateness Measure (IAM) mean ranged from 3.00 to 3.96, reaching its lowest value in June before recovering to 3.75 in August.

Qualitative Study

Study Design

A qualitative descriptive study using semi-structured interviews was undertaken to understand the experiences, perceptions, and implementation needs of key stakeholders involved in ICU end-of-life care.

The qualitative work was embedded within the implementation study and informed iterative refinement of the MCACP.

Participant Sampling

Purposive sampling was used to ensure representation from all stakeholder groups involved in ICU decision-making.

Participants included:

- ICU consultants
- critical care fellows
- resident doctors
- ICU nurses
- palliative care physicians

- hospital administrators
- family caregivers
- implementation team members

Sampling continued until thematic saturation was achieved within stakeholder groups.

Data Collection

Semi-structured interview guides were developed using findings from the six realist reviews and aligned with the Consolidated Framework for Implementation Research (CFIR).

Separate interview guides were prepared for clinicians, nurses, administrators, and family caregivers.

Interviews explored:

- current ICU practices
- communication experiences
- prognostication
- shared decision-making
- treatment limitation
- legal concerns
- organisational culture
- workflow barriers
- multidisciplinary collaboration
- acceptability of MCACP
- recommendations for improvement

Interviews were conducted by trained qualitative researchers employed under the CAR programme.

Prior to commencement, interviewers underwent training in qualitative interviewing, reflexivity, ethical research conduct, and realist interviewing techniques.

Each interview lasted approximately 45–75 minutes.

All interviews were:

- audio-recorded
- transcribed verbatim
- anonymised
- verified for accuracy

- stored in password-protected databases

Field notes documenting contextual observations, interviewer reflections, and implementation issues were maintained after every interview.

Qualitative Data Analysis

Qualitative analysis followed an iterative thematic analysis approach.

The analysis consisted of:

Step 1

Repeated reading of transcripts for familiarisation.

Step 2

Line-by-line open coding.

Step 3

Development of the initial codebook.

Step 4

Independent coding by two researchers.

Step 5

Consensus meetings to resolve coding differences.

Step 6

Grouping of related codes into categories.

Step 7

Generation of overarching themes.

Coding was informed by both inductive thematic analysis and deductive implementation science constructs derived from CFIR and the realist programme theory.

Example Coding Framework

Participant Quote	Initial Code	Category	Theme
"We usually wait until families ask..."	Delayed prognosis discussion	Communication practices	Delayed goals-of-care discussions
"Different consultants often have different opinions."	Variable physician opinions	Decision-making	Lack of multidisciplinary consensus
"Documentation is different	Inconsistent	Organisational	Lack of standardisation

Participant Quote	Initial Code	Category	Theme
for every consultant."	documentation	workflow	
"Families expect us to continue everything."	Unrealistic expectations	Family engagement	Communication and shared decision-making

Preliminary Themes

Thematic analysis identified several recurring themes:

- Prognostic uncertainty delays initiation of appropriate care.
- Communication with families remains largely unstructured.
- Decisions are influenced by multiple clinicians with limited formal consensus mechanisms.
- Documentation of goals of care is inconsistent.
- Specialist palliative care involvement occurs late in the illness trajectory.
- Clinicians expressed a strong desire for structured communication pathways and practical implementation guidance.
- Existing ICU workflows require intervention components that are simple, flexible, and integrated into routine practice.

Integration of Quantitative and Qualitative Findings

Quantitative and qualitative findings were reviewed jointly during multidisciplinary investigator meetings.

The quantitative data demonstrated low rates of discussions of prognosis, family meetings, advance care planning, and palliative care referrals despite high symptom burden.

The qualitative interviews explained these findings by identifying organisational barriers, prognostic uncertainty, workflow limitations, documentation challenges, legal concerns, and variability in multidisciplinary decision-making.

This triangulation directly informed refinement of the MCACP intervention, particularly the development of prognostication triggers, communication guides, multidisciplinary physician review pathways, documentation templates, and implementation workflows.

Planned Publications

Baseline ICU palliative care practices and implementation readiness in two Indian tertiary ICUs: A mixed-methods study.

Work Package 2.2: Pilot Implementation and Workflow Optimisation

Methodology

The preliminary MCACP intervention is being implemented using an iterative co-development approach. Implementation focuses on integrating the seven MCACP components into existing ICU workflows while identifying operational challenges, implementation barriers, and opportunities for adaptation. Continuous learning cycles involving frontline clinicians, nurses, implementation champions, and palliative care specialists are undertaken to optimise delivery across both open and closed ICU models.

Data Collection and Analysis

Implementation outcomes include intervention fidelity, acceptability, feasibility, adoption, usability, workflow integration, staff engagement, and training completion. Data sources include implementation logs, structured observations, field notes, workflow mapping, staff feedback, and qualitative interviews. Qualitative findings are analysed thematically using CFIR and NPT constructs to understand mechanisms that facilitate or hinder successful implementation.

Field observations: To understand the ICU setting, we observed the ICU workflow, admission process, billing process, counselling, bedside nursing care, transitions, and medical board meetings. A framework to record observational field notes was prepared for this.

Preliminary Results

Implementation activities are ongoing at KMC Manipal, with workflow testing and refinement of intervention components. At St. John's Medical College, implementation workshops have commenced following site initiation. Early observations indicate the need for simplified documentation, clearly defined referral triggers, structured multidisciplinary family meetings, and workflow integration strategies that minimise disruption to routine ICU practice while improving consistency of appropriate care delivery.

Planned Publications

Implementation of the Multifaceted Comprehensive Appropriate Care Package (MCACP): Early implementation experiences from two pilot ICUs in India.

Work Package 2.3: Refinement of the MCACP Intervention and Manual Development Methodology

Findings from the six realist reviews, baseline assessments, implementation experience, workflow observations, and stakeholder feedback will be synthesised to iteratively refine the MCACP intervention. Human-centred design methods, Delphi consensus, and stakeholder workshops will be used to ensure the intervention remains context-sensitive, feasible, and scalable. Refinements will be documented using the TIDieR framework to facilitate reproducibility and future implementation.

Data Collection and Analysis

Quantitative implementation metrics and qualitative stakeholder feedback will be triangulated to identify intervention components requiring modification. Consensus meetings will determine the core intervention elements, optional context-specific adaptations, implementation tools, educational resources, and quality indicators to be incorporated into the final MCACP manual.

Preliminary Results

The pilot work has confirmed several priority areas requiring refinement, including structured prognostication pathways, multidisciplinary physician consensus mechanisms, standardised communication guides, shared decision-making tools, comprehensive anticipatory care planning templates, symptom management algorithms, and care evaluation indicators. These findings are directly informing the development of the first draft of the MCACP manual and associated implementation toolkit.

Planned Publications

1. *Development of the MCACP intervention manual using realist evidence and pilot implementation findings.*
2. *Co-design and refinement of a comprehensive ICU palliative care intervention for India: A mixed-methods implementation study.*

Publications -

Vallath N, Parvathy V, Padhi P, Johny T A, Krishnakumar M, Varghese PS. Procedures to implement the Supreme Court of India directives to withdraw/withhold life-sustaining medical treatment. Indian J Med Res. 2026 Mar;163(3):319-326. doi: 10.25259/IJMR_3005_2025. PMID: 42024905; PMCID: PMC13126532.

Quality Improvement project: addressing quality gaps in the documentation of medical certificates of cause of death in the palliative medicine department, IP setting, to improve patient care: led by Dr Angeline (DNB Palliative medicine) and mentored by Dr Nandini Vallath. Drafting of the manuscript is ongoing.

A Quality Improvement (QI) study was undertaken to reduce the Discharge Against Medical Advice (DAMA) rate through collaboration between the Department of Pain and Palliative Medicine and Supportive Care, the Critical Care Department, and the Surgical and Neuro Critical Care Departments. The study identified key drivers contributing to DAMA and implemented strategies to address them. The findings and key drivers from this QI initiative will be incorporated into the ongoing ICMR study to inform interventions to reduce DAMA rates and improve continuity of care for critically ill patients transitioning to palliative care (under preparation for publication).

Presentation in conference

- Dr. Amarja Havaladar, presented paper on Quality improvement study to reduce incidence of discharge against medical advice (DAMA) from intensive care units of a tertiary academic centre, using the A3 methodological approach in *Criticare 2026*
- Dr. Angelin Johny, presented paper on Determinants of discharge against medical advice (DAMA) in ICU: A prospective quality improvement study on modifiable contributors in 33rd *IAPCON 2026*.

Project Personnel and Responsibilities During the Reporting Period

Kasturba Medical College (KMC), Manipal (12 Personnel)

1. Site Implementation Lead Support (1)

The Site Implementation Lead Support coordinated the day-to-day implementation of the CAR project at KMC. Responsibilities included coordinating study activities across the ICU and palliative care teams, monitoring adherence to study timelines, organising investigator meetings, supervising recruitment activities, resolving operational issues, facilitating implementation of the MCACP intervention, and ensuring communication between the principal investigators and research staff. The role also involved supporting workflow

integration, documenting implementation challenges, and coordinating iterative refinement of intervention components.

2. Training Programme Coordinator (1)

The Training Programme Coordinator was responsible for planning, organising, and delivering educational activities related to the MCACP intervention. Responsibilities included preparing training schedules, developing educational materials, organising orientation programmes for clinicians and research staff, maintaining attendance records, coordinating simulation sessions where applicable, evaluating training outcomes, and ensuring the competency of staff involved in intervention delivery.

3. Data Management and Reporting Coordinator (1)

The Data Management and Reporting Coordinator established and maintained the project database and ensured data quality throughout the study. Responsibilities included database development, preparation of electronic case report forms, data validation, quality assurance checks, data cleaning, generation of recruitment reports, preparation of progress reports, maintenance of study documentation, and supporting interim analyses. The coordinator also monitored data completeness across study sites and prepared reports for investigators, ethics committees, and the funding agency.

4. Qualitative Research Coordinator (1)

The Qualitative Research Coordinator led all qualitative components of the study. Responsibilities included developing interview guides, scheduling interviews and focus group discussions, training interviewers, supervising qualitative data collection, managing transcription, quality-assuring transcripts, supervising coding, conducting thematic analysis, maintaining qualitative databases, and preparing qualitative summaries to inform intervention refinement.

5. Digital Content Liaison (1)

The Digital Content Liaison developed and managed educational and communication resources required for implementation. Responsibilities included preparing digital learning materials, managing training videos, updating online educational resources, coordinating

dissemination materials, maintaining project communication platforms, supporting webinar organisation, and assisting with digital documentation of project activities.

6. Liaison for Multi-site Communication (1)

This role ensured effective coordination between participating institutions. Responsibilities included scheduling regular multicentre meetings, maintaining communication between investigators, tracking project milestones across sites, coordinating document sharing, preparing meeting minutes, monitoring action points, facilitating collaborative decision-making, and supporting harmonisation of study procedures across centres.

7. Regulatory and Ethics Documentation Officer (1)

The Regulatory and Ethics Documentation Officer coordinated all regulatory activities. Responsibilities included preparing and submitting ethics applications, maintaining regulatory files, tracking approvals and renewals, ensuring compliance with institutional and national regulations, preparing protocol amendments, maintaining investigator site files, documenting adverse events where applicable, and ensuring adherence to Good Clinical Practice (GCP) documentation standards.

8. Project Financial and Administrative Support (1)

The Project Financial and Administrative Support Officer managed the administrative aspects of the CAR grant. Responsibilities included financial documentation, budget monitoring, procurement of study materials, processing reimbursements, maintaining expenditure records, coordinating travel and logistics for project meetings, supporting human resource documentation, and ensuring compliance with institutional financial procedures and ICMR reporting requirements.

9. Project Research Scientist – Implementation (1)

The Project Research Scientist (Implementation) coordinated the implementation of the MCACP intervention at the clinical level. Responsibilities included supporting participant recruitment, supervising intervention delivery, monitoring implementation fidelity, conducting workflow observations, facilitating multidisciplinary meetings, documenting

implementation barriers and facilitators, supporting clinician engagement, and contributing to refinement of intervention components based on implementation experience.

10. Project Research Scientist – Evaluation (1)

The Project Research Scientist (Evaluation) was responsible for monitoring study outcomes and evaluating implementation processes. Responsibilities included oversight of quantitative data collection, quality assurance of clinical data, statistical analyses, implementation outcome evaluation (feasibility, acceptability, fidelity, adoption), integration of quantitative and qualitative findings, preparation of reports, and contribution to manuscript development.

11. Medical Social Worker (MSW) (1)

The Medical Social Worker supported participant engagement and the project's family-centred aspects. Responsibilities included identifying eligible family participants, coordinating informed consent, facilitating qualitative interviews with caregivers, supporting family meetings where appropriate, documenting psychosocial concerns, assisting with follow-up assessments, and acting as a liaison between research staff and families during data collection.

12. Office Helper (1)

The Office Helper provided logistical and administrative support essential for the smooth conduct of project activities. Responsibilities included organising study materials, maintaining project records, supporting meeting logistics, assisting with document management, facilitating the movement of study documents between departments, coordinating office supplies, and supporting day-to-day administrative operations.

St. John's Medical College (SJMC), Bengaluru (4 Personnel)

1. Project Research Scientist – Implementation (1)

The Project Research Scientist coordinated the implementation of the MCACP intervention at SJMC. Responsibilities included participant screening and recruitment, supervision of intervention delivery, coordination with ICU clinicians, workflow mapping, implementation

monitoring, documentation of implementation challenges, and participation in regular multicentre review meetings.

2. Project Research Scientist – Evaluation (1)

The Project Research Scientist (Evaluation) oversaw quantitative and implementation evaluation activities. Responsibilities included data quality monitoring, participant follow-up, implementation outcome assessment, statistical analyses, preparation of interim reports, and integration of evaluation findings into intervention refinement.

3. Research Psychologist (1)

The Research Psychologist led the psychosocial and qualitative components at SJMC. Responsibilities included conducting semi-structured interviews with family caregivers and healthcare professionals; assessing communication experiences; exploring psychological and emotional aspects of end-of-life care; facilitating qualitative data collection; assisting with coding and thematic analysis; and contributing to the development of the MCACP's communication and family support components.

4. Research Nurse (1)

The Research Nurse supported clinical implementation and participant recruitment within the ICU. Responsibilities included daily screening of eligible patients, collection of clinical and nursing data, maintenance of study documentation, coordination of informed consent procedures, monitoring adherence to the study protocol, assisting with symptom assessments, facilitating communication between the clinical and research teams, and ensuring completeness of bedside data collection.

Annexure 3 – online training module development

Background

Appropriate EOLC and palliative care for patients coming to the ICU are now established practices worldwide. This helps avoid inappropriate life-sustaining treatments in the ICU and mitigates suffering for patients and their families.

In India, there has been considerable progress over the last 20 years in developing and refining EOLC practice guidelines through professional societies such as ISCCM and IAPC. Patient advocacy and advocacy by these societies have helped frame landmark legal judgments that lend vital support to these guidelines. Despite this, uptake of these guidelines has been patchy and clinical practice at the ground level has not changed much. ICMR has allocated funds to establish a Centre for Advanced Research (CAR), primarily tasked with developing and implementing a Multifaceted Comprehensive Appropriate Care Package (MACACP) nationwide.

As part of this project and in collaboration with ISCCM, an EOLC & Palliative Care ICU Course (EPIC) is planned.

Aim

To create and implement an EOLC & Palliative Care ICU Course (EPIC) that is focused on training intensivists in India.

Objectives

1. Create a competency-based curriculum for EPIC based on the best available evidence that is in consonance with current practice and legal requirements
2. Prepare a teaching program that is interactive and has an online and offline component. The online component will have 12 modules (each with 2 sessions) spread over 3-4 months. The offline component will be conducted over 2 days and will have a strong focus on clinical ethical reasoning and communication skills

Session	Sr. No.	Modules	Dates	TOPIC
---------	---------	---------	-------	-------

1	1	What is Palliative Care? Why Does it Matter? Why palliative care in ICU?		Define palliative care (PC)
				Principles of PC – what, why, when, how
				Evaluate myths and assumptions about PC
				Present Rationale and Data for integrating PC in the ICU
				Describe when and how to utilise a subspecialty palliative care team (models for ICU-PC Integration)
	2	What is end-of-life care		Define EOLC (various definitions)
				Identifying objectives of good EOLC
				Principles and components of a good death
				Identifying palliative care needs in the ICU – Screening and Trigger Criteria for PC in the ICU
				Rapid Palliative care assessment in the ICU
2	3	Illness Trajectories and Prognostication		Nutrition/Hydration at EOL
				Explore the five trajectories of Dying
				Identifying palliative care needs through the illness trajectories
				Prognostication in chronic illnesses
				Disease-specific prognostic tools
	4	Ethical & Legal Issues in EOLC		Performance status measures
				Recognising futility
				Legal policies at the end of life care
	5	Basic communication		Policies and guidelines for withdrawing and withholding life-sustaining treatment
				Physician-assisted suicide
				Euthanasia
	5	Basic communication		Ethical issues in end of life care
				Evaluating capacity and identifying decision-makers in the ICU
	5	Basic communication		Learn methods for communicating prognosis
				Six steps of the SPIKES model for BBN
				SOLER/NURSE model

3	6	Complex		Advance care plan/advance directives
		Communicati		Withdrawing and withholding life support
		on Situations		DNR/DNI
				Anger and Conflict Management
				Denial and Collusion
4	7	Goals of care discussion and decision-making		Understand ways to elicit goals of care
				Eliciting and resolving concerns at the end of life care decision making (Consensus building)
				Steps in shared-decision making
	8	Handling difficult families		Types of families
				The principles of the family meeting in the ICU
				Using 'VALUE' in family communications
				Addressing conflicts within families
	5	9	Patient and family-centred care	
				Learn the four critical components of PFCC
				Describe strategies for delivering effective PFCC in palliative care
				Understand and reduce barriers to the delivery of PFCC
10		Assessment and Management of Pain		Pathophysiology and types of pain
				Need for pain assessment in the ICU
				Assessment of pain in the ICU
				Subjective self-report assessment tools
				Objective behavioural assessment tools
				Pain Management in the ICU
		Non-pharmacological management		
		Pharmacological management		
	11	Assessment and Management of Dyspnoea		Pathophysiology of dyspnoea in the terminally ill
				Quantification of dyspnoea
				Intensity
				Severity
				Impact
		Management of dyspnoea in the ICU		

6				Pharmacological management – opioids, anxiolytics, oxygen therapy Non-pharmacological management
	12	Assessment and Management of Respiratory Secretions/Thirst/Xerostomia		Evidence-based recommendations for airway hygiene in the ICU
				Assessment of respiratory secretions
				Management of respiratory secretions Pharmacological management Non-pharmacological management
				Management of thirst and xerostomia
7	13	Assessment and Management of Delirium/Agitation/Restlessness		Causes and types of delirium in ICU
				Assessment of delirium
				Management of delirium Pharmacological management Non-pharmacological management
	14	Management of psychospiritual distress in the ICU		Describe screening for psychospiritual pain in the ICU
				Describe approaches to managing spiritual and emotional distress of patients and their caregivers
				Identify thresholds for referral to mental health clinicians
8	15	Management of Refractory Distress		Describe persistent myths about palliative sedation
				Explain the circumstances under which palliative sedation may be indicated
				Describe recommended dosing for palliative sedation
				Review the potential alternatives to palliative sedation
	16	Grief and Bereavement		Review theories of grief
				Grief in sudden death
				Assess grief in patients/family members
				Discuss strategies to address grief in patients/families

				Principles of bereavement support to families
9	17	Devices and discontinuation of devices		Overview of cardiac rhythm management devices
				Deactivation of implantable cardioverter defibrillators and pacemakers – differences and challenges
				Logistics of deactivation
				Modes of deactivation
				Ethical and legal concerns in the deactivation
				Communication issues
	18	Organ donation and body donation		WHO guiding principles on human cell, tissue, organ transplantation
				Legal and ethical aspects of organ and body donations
				Effective communication in organ donation
10	19	Discharge against medical advise		Prevalence of DAMA
				Causes and factors influencing DAMA
				Consequences of DAMA
				Ethical aspects of DAMA
				Legal aspects of DAMA
	20	Self-Care for Professionals		Understand how self-care is a core competency in palliative and end of life care
				Recognise what triggers stress and burnout
				Develop a self-care plan that reduces stress and fosters personal growth and well-being
11	21	Team Collaboration and Effectiveness		Identify the conceptual basis for teamwork in ICU palliative care
				Describe different types of teams in palliative care
				Name specific advantages and challenges of teamwork
				Principles of Intra-team and Inter-team communication in ICU-PC
	22	Introducing Quality Improvement		Discuss why a quality improvement methodology is an essential approach for integrating palliative care into ICU services
				Describe the basic concepts of quality improvement

12		in ICU-PC		approaches
				Describe the first steps to improving quality
				Complete a performance improvement project
	23	PC in Paediatric ICU		Define paediatric palliative care (PPC)
				Learn how children of different developmental capacity understand concepts of illness and death and how this can impact care planning
				Use developmentally based strategies to identify and address grief in children
				Discuss grief related to the loss of a child and strategies to address family grief
				Discuss symptom management in children
	24	PC in Perinatal/Neonatal ICU		Define neonatal palliative care and who need it
				Principles of symptom management in neonates Pain Respiratory distress/secretions Agitation Compassionate extubation Anticipatory guidance for lactating mothers
				Addressing existential distress and psychosocial care
				Addressing bereavement in families

PROGRAM OVERVIEW

Title

“EOLC & Palliative Care ICU Course (EPIC)”

- Duration: 6 months
- Online sessions: Every 2 weeks (2 hours each)
- Structure per session:
 - Interactive lecture – 30-45 min
 - Small group discussion (breakout sessions) – 45-60 min

Module 1.

- Introduction to course - 10 min
- Small group breakout discussion (core learning) - 45 min
 - Discuss the terms palliative care, EOLC, Passive euthanasia, Euthanasia, Chronic illness, terminal illness
- Talk 1 – What is palliative care why does it matter in ICU - 30 min
- Talk 2 - What is end-of-life care and what are the different related terms - 30 min

Module 2.

- Illness Trajectories and Prognostication
- Ethical & Legal Issues in EOLC
- Small group breakout discussion (core learning) - 45 min

Module 3.

Basic communication

Complex Communication Situations

Small group breakout discussion (core learning) - 45 min

Module 4.

- Goals of care discussion and decision-making
- Handling difficult families
- Small group breakout discussion (core learning) - 45 min

Module 5.

- Patient and family-centred care
- Assessment and Management of Pain
- Small group breakout discussion (core learning) - 45 min

Module 6.

- Assessment and Management of Dyspnoea
- Assessment and Management of Respiratory Secretions/Thirst/Xerostomia
- Small group breakout discussion (core learning) - 45 min

Module 7.

- Assessment and Management of Delirium/Agitation/Restlessness
- Management of psychospiritual distress in the ICU
- Small group breakout discussion (core learning) - 45 min

Module 8.

- Management of Refractory Distress
- Grief and Bereavement
- Small group breakout discussion (core learning) - 45 min

Module 9.

- Devices and discontinuation of devices
- Organ donation and body donation
- Small group breakout discussion (core learning) - 45 min

Module 10.

- Discharge against medical advise
- Self-Care for Professionals
- Small group breakout discussion (core learning) - 45 min

Module 11.

- Team Collaboration and Effectiveness
- Introducing Quality Improvement in ICU-PC
- Small group breakout discussion (core learning) - 45 min

Module 12.

- PC in Paediatric ICU

- PC in Perinatal/Neonatal ICU
- Small group breakout discussion (core learning) - 45 min
- Final Component:
2-day in-person contact session (skills-heavy)

BREAKOUT DESIGN (THIS IS YOUR PROGRAM'S STRENGTH)

Each breakout group (6–8 participants):

- 1 facilitator
- 1 case scenario
- Structured worksheet:
 - What is happening?
 - What are your concerns?
 - What would you say/do?
 - What are the ethical/legal issues?

End with 2–3 key takeaways per group

2-DAY CONTACT SESSION (CORE SKILLS LAB)

Focus: Communication Mastery + Real ICU Scenarios

Day 1: Foundations of Communication

Modules:

- Breaking bad news (SPIKES in ICU context)
- Discussing prognosis & uncertainty
- Handling emotions (anger, denial, silence)

Methods:

- Demonstration videos
- Role play with feedback
- Faculty modeling

Day 2: Advanced Conversations

Modules:

- Goals-of-care discussions
- DNAR & withdrawal conversations
- Conflict resolution (family vs team)
- Death communication & bereavement

Methods:

- Simulated ICU scenarios
- Standardized patients (if possible)
- Video recording + feedback

ASSESSMENT STRATEGY

During Online Program:

- Participation in breakout discussions
- Reflective notes (short, optional)

During Contact Session:

- Mini-CEX (communication scenarios)
- Faculty feedback

Optional:

- Pre-post confidence/self-efficacy scale

OUTPUTS FOR PARTICIPANTS

- ICU EOLC checklist toolkit
- Sample documentation templates (India-specific)
- Communication phrase bank
- Case compendium

Competency based curriculum

Competencies linked to EOLC steps

EOLC Step	Core Competen	Learning Objectives	Key Topics	Skills (Observable)	Teaching-Learning
-----------	---------------	---------------------	------------	---------------------	-------------------

	cy	(Learner will be able to...)			Methods
1. Recognize dying patient	Clinical recognition & prognostication	Identify patients with poor prognosis and likely dying trajectory	Prognostication, medical futility, illness trajectories	Identify EOLC candidates, interpret clinical course	Bedside teaching, case-based discussions (CBD), mortality reviews
2. Initiate EOLC discussion	Communication initiation	Initiate timely and appropriate EOLC conversations	Breaking bad news, uncertainty communication	Start goals-of-care discussion, use SPIKES	Role play, simulation, observed encounters
3. Assess preferences & values	Respect for autonomy & cultural competence	Elicit patient/family values and preferences	Advance care planning, surrogate decision-making, cultural context	Identify surrogate, explore values, document preferences	Bedside interviews, reflective exercises, small group discussions
4. Conduct family meeting	Family-centered communication	Conduct structured multidisciplinary family meetings	Family dynamics, conflict resolution, empathy	Lead family meeting, manage emotions, summarize plan	Simulation, video-based learning, Mini-CEX
5. Achieve consensus on goals of care	Shared decision-making & ethics	Facilitate consensus aligning medical and family perspectives	Ethical principles (autonomy, beneficence), shared decision-making	Negotiate goals, resolve disagreement, document consensus	Ethics case discussions, facilitated group dialogue

6. Decide withholding/withdrawal	Ethical & clinical decision-making	Apply ethical reasoning to limit life-sustaining treatment	Withholding vs withdrawal, futility, proportionality	Recommend limitation of treatment, justify decisions	Case-based ethics teaching, OSCE scenarios
7. Legal documentation	Legal literacy & compliance	Document EOLC decisions as per Indian legal framework	DNAR, Supreme Court guidelines, PMB/SMB process	Complete forms, maintain documentation, ensure consent	Workshops, document audits, supervised practice
8. Implement EOLC plan	Systems-based practice & teamwork	Translate decisions into actionable ICU care plans	ICU protocols, deprescribing, team coordination	Write EOLC orders, coordinate with nursing team	Bedside supervision, protocol-based teaching
9. Symptom control & comfort care	Clinical palliative care	Manage end-of-life symptoms effectively	Pain, dyspnea, delirium, palliative sedation	Prescribe opioids/sedatives, titrate comfort measures	Bedside teaching, skill stations, case discussions
10. End-of-life support & after-death care	Compassionate care & bereavement support	Provide dignified death and support family post-death	Death communication, bereavement, rituals	Declare death sensitively, support family, conduct debrief	Role play, reflective practice, debrief sessions

Cross-Cutting Competencies

Competency	Learning Objectives	Teaching Methods
Communication mastery	Demonstrate empathy, clarity, and structured communication	Simulation, role play, Mini-CEX
Ethical reasoning	Apply ethical principles in ICU decisions	Ethics rounds, case discussions
Legal awareness (India)	Practice within Indian legal framework	Workshops, policy teaching
Cultural & spiritual sensitivity	Respect diverse beliefs and practices	Narrative medicine, reflective writing
Teamwork & leadership	Lead interdisciplinary care	Team-based simulation
Professionalism & self-care	Manage moral distress and burnout	Reflective sessions, peer discussion

Annexure 4 – training workshop conducted at SJMC, Bengaluru

SJMC Workshops –

Qualitative research workshop

A qualitative research methods training workshop series was conducted over a three-week period on 28 February, 7 March, and 14 March by the Pain and Palliative department, SJMCH. The workshops were facilitated by experts in qualitative research methodology and focused on strengthening research competencies relevant to the project. Domain experts included those from the St. John's Research Institute (SJRI), Bengaluru; the National Institute of Mental Health & Neurosciences (NIMHANS), Bengaluru; and the University of Pittsburgh, USA. A total of 25 participants were included, comprising clinicians, residents, researchers from SJMCH and SJRI, researchers from NIMHANS, and researchers from the Swami Vivekananda Youth Movement (SVYM). The workshop covered lecture, discussion, and hands-on training on

- Orientation to qualitative research paradigm
- Where do we fit in the qualitative methods in critical care, and what approaches to choose from?
- Crafting a qualitative research protocol
- Connecting the dots: Insights from implementing Qualitative research in Critical Care
- Voices from the field: data collection (Interviews and focus group discussions)
- From field to findings: rigour and ethics in qualitative inquiry
- Thematic analysis in action (Coding, naming and defining themes, category development)
- Digital tools for qualitative research (Introduction to the digital tools and hands-on training on NVivo coding)

Implementation research workshop

Objectives

When participants return to their ICUs after the workshop, what are we hoping they will do?

1. A shared understanding of the evidence and policies for palliative care in the ICU
2. Reflect on the implementation gaps and strategies to address it.
3. Teams/participants go back with a problem statement/theme they can work on. Including team/stakeholders, measurement & project/study design.

Note: Room layout - movable chairs and tables for group work. Avoid auditoriums with fixed seats.

Attendees: Ideally 3-4 people from the same unit, including nurses, intensivists, MDT palliative care personnel, and others who are involved.

Agenda

Time	Topic	Session Format	Speaker
Day 1, Friday, June 26			
2:00 - 2:30 PM	Welcome and housekeeping		Dr Vallath – St Johns Dr Seth – Univ Pittsburgh
	Intros, motivations, teams	Speed Networking - 3 rounds - 3 mins each. Conversation points - What aspect of life makes today worthwhile? - What's something you did recently, at work or outside work, that left you feeling happy or satisfied & what is one <i>specific</i> thing you wish you could magically improve for patients in your unit?	
2:30 - 2:45 PM	ICMR opportunities and current work in Implementation science	Plenary – Online	Dr Neha Dahiya, Dr Ashoo Grover (ICMR)
2:45 - 3:00 PM	How has policy and implementation science worked for the greater good - lessons from prior successes (& failures)	Plenary - Online	Dr Swarup Sarkar, former WHO SEARO Director
10 min break	Tea		
3:15 - 3:35 PM	Improvement basics, examples & how it fits	Plenary - guidance for group work. teams & facilitators	Dr Vaid --WHOCC, Delhi

	with implementation science Guidance for group work	assigned.	
3:35PM - 4:45 PM	Identifying the problem - teams make a list of opportunities for improvement and rank by impact on patient outcomes, feasibility, resources, acceptability etc. (Assign sub-themes or they pick what they like?)	Team Studio Group Work 1	Dr P Chatterjee Dr Vaid-WHOCC, Delhi Dr A Chatterjee Dr Bhavna / Dr Vaid
4:45 - 5 PM	Report back	One thing that you heard that you'd like to share - max 3-4 people	Bhavna, N Vallath
Day 2, Saturday, June 27			
12:00 - 12:15 PM	Recap prior day, Qs	Speed Networking - 3 rounds - Something new I learnt yesterday - Something that I connect strongly with - Something I don't understand (other person tries to help)	Dr Pratyasa, Dr Vaid
12:15 - 12:45 PM	CFIR framework: Understanding systems and barriers & facilitators	Plenary Followed by 15 min Q & A	Drs Parvathy/Garima
12:45 - 1:30 PM	Lunch break		

1:30 -2:30 PM	Expert panel, current policy change, foreseeable facilitators & barriers. Policy/upstream factors as barriers/facilitators to pall care adoption, how research may help +- solve	Panel moderated by Dr N Salins	ICU/ Pall care experts Dr Bhuvana, Dr Anuja, Dr Rajesh Shetty, Assigned Intensivists, CMS, ANS critical care
2:40 PM - 3 PM	Outcomes and measurement, phases: adoption, fidelity, feasibility, acceptability, appropriateness, penetration, sustainability	Plenary	Dr Prem Mony, SJRI
2:30 - 2:40 PM	Break 10 min		
3:00 - 4:00 pm	Applying CFIR to an implementation question/ project theme, based on what was discussed	Team Studio Group Work 2 (Posters on wall with specific themes, people go work on the theme assigned List strategies on slide	Facilitators, Moderators Dr Bhuvana/Dr Vallath/Dr Mony/Assigned intensivists/ Garima/ Dr Arun
4PM - 5PM	Pall care Impl Sci US	Plenary	Dr Gina Piscitello, University of Pittsburgh, (online)
Day 3, Sunday, June 28			
9 AM - 9:30AM	Study example in use - colon cancer screening adoption RCT	A study applying the principles learnt the previous day	Dr Divya Shankar- Boston University(online)
10:15 AM -	Implementation	Plenary	Dr Parvathy/Dr Pratyasha

10:30 AM	strategies: ERIC taxonomy, tailoring, and strategy specification		
10:30 - 11 AM	Discuss possible measures/ available data for integrating palliative care in ICU. What is available? What is not.	Team Studio Group Work 3 (using project ideas/themes list available data)	
11 AM - 11:20 AM	Implications of policy research on implementation of operationalisation of evidence - especially in the Indian context	Plenary	Dr Anirban Chatterjee - University of South Caroline
11:20AM - 11:30AM	What makes an implementation idea publishable	Plenary	Dr Nagi – Publishing - Springer Nature Journal
10 minutes	Break		
11:40 - 12:30	Team Studio: top determinants, strategies, outcomes and priorities	Team Studio Group Work 4 - interactive format - TRIZ 1) What we need to ‘stop’ doing 2) Commitments we can make	Dr Vaid, former WHO
12:30 - 1PM	Feedback, suggestions, Validictory, Certificate announcement		
Lunch			

Documents / administrative changes developed for introducing PC upstream

Non-escalation-related	SJMC FORM 1 screening and enrolment baseline and admission characteristics initial care needs and clinical status the pathway to non-escalation, patient/family confidence and comfort with the decision. FORM 2 the quality of inpatient palliative care the processes and support involved in transitioning patients to their preferred place of care – if any continuity of care and patient outcomes during follow-up beyond discharge. FORM 3 family satisfaction, understanding, preparedness, decision-making experience, emotional responses Experience of opting for non-escalation, consequences. FORM 4 Direct and indirect costs healthcare resource utilization financial impact of non-escalation and palliative care on patients and families. KMC – the BLUE MAPLE document
Goals of care document/processes	SJMC Constitution of the primary, secondary medical boards (PMB, SMB) and reporting formats for PMB, SMB evaluations has been accomplished KMC – Critical care review board
living will worksheet template	SJMC 1.SOPs for Transitioning ICU patient with guarded prognosis to Palliative Care ApTCare-ICU-Criteria for Appropriate Transitions in Care for person with serious illness - Applied at / after 2 days of ICU 2.SOPs for compassionate extubating and EOLC within the ICU

Annexure 5 - publications arising from the project.

Published x 1

Ghoshal A, Prasad A, Nanda A, *et al* How prognostic information influences care planning in adult intensive care units: protocol for a realist review. *BMJ Open* 2026;**16**:e117449. doi: 10.1136/bmjopen-2026-117449

Publication history

Received January 27, 2026
Accepted May 29, 2026
First published June 29, 2026.
Online issue publication June 29, 2026

in review x 5

A Realist Review Protocol of Communication Practices in Intensive Care Units

STATUS	ID	TITLE	CREATED	SUBMITTED
✉ Contact Journal EPA: Editorial Office, BMJ Open	bmjopen-2026-123169	A Realist Review Protocol of Communication Practices in Intensive Care Units View Submission [View Original Files] Cover Letter	25-May-2026	28-May-2026
Awaiting PRM Assignment				

Advance Care Planning in Intensive Care Units for People with Serious Life-Limiting Illnesses: A Realist Scoping Review Protocol

✉ Contact Journal EPA: Editorial Office, BMJ Open	bmjopen-2026-123179	Advance Care Planning in Intensive Care Units for People with Serious Life-Limiting Illnesses: A Realist Scoping Review Protocol View Submission [View Original Files] Cover Letter	25-May-2026	25-May-2026
Awaiting PRM Assignment				

Protocol: Realist Review of Shared Decision Making in Intensive Care Units

✉ Contact Journal EPA: Editorial Office, BMJ Open	bmjopen-2026-120758	Protocol: Realist Review of Shared Decision Making in Intensive Care Units View Submission [View Original Files] Cover Letter	06-Apr-2026	06-Apr-2026
Awaiting ME Decision				

Structured ICU Governance Mechanisms to Strengthen End-of-Life Decision-Making: A Realist Review Protocol

✉ Contact Journal	bmjopen-2026-119520	Structured ICU Governance Mechanisms to Strengthen End-of-Life Decision-Making: A Realist Review Protocol	12-Mar-2026	12-Mar-2026
EPA: Editorial Office, BMJ Open		View Submission [View Original Files]		
• Awaiting Reviewer Reply		Cover Letter		

 **SNAPP**

BMC Health Services Research

 [Notifications](#) ²

 [Account](#)

Implementation of a Multifaceted, Comprehensive, and Suitable Care Package for Patients with Advanced

CURRENT STATUS

Your revised submission has passed our technical checks

Your revised submission has passed the technical checks and is with the editor that will review before making a decision or send it for peer review.

The length of time this takes can depend on several factors, including finding the best peer reviewers for your submission.

Progress so far [Show history](#)

Submission received

Technical check

Editorial assignment

With editor

Under preparation x 3

- Advance Care Planning in Intensive Care Units for People with Serious Life-Limiting Illnesses: A Realist Scoping Review
- Realist Review of Shared Decision Making in Intensive Care Units
- Structured ICU Governance Mechanisms to Strengthen End-of-Life Decision-Making: A Realist Review

85

Annexure 5 - Initial MCACP Package, Version 1.0, dated 31 August 2026.

Document control	Details
Version	1.0 - initial co-developed package
Version date	31 August 2026
Programme	MATHRU
Principal Investigator	Dr Naveen Salins
Pilot sites	KMC Manipal and St John's Medical College Hospital, Bengaluru
Status	Controlled working version for pilot implementation and formal consensus/usability completion

Important status statement: This document freezes the initial package available at the Year-1 reporting cut-off. It does not represent final Objective-1 closure. Protocol-specified two-step Delphi consensus, separate documentation of the NGT-informed process, final TIDieR reporting, and formal frontline usability testing remain to be completed and linked to the next controlled version.

Document purpose and contents

This manual defines the initial MCACP intervention, its core functions, minimum delivery standards, adaptable operational forms, clinical and communication workflows, implementation resources, and evaluation framework. It is intended for pilot use in Indian adult ICUs with different organisational models and levels of specialist palliative-care integration.

Contents register

- 1. Scope, target population and principles
- 2. Development basis and initial programme theory
- 3. Entry criteria and identification workflow
- 4. Seven MCACP components
- 5. Integrated clinical workflow
- 6. Team roles and governance
- 7. Implementation tools and documentation templates
- 8. Training and competency package
- 9. Fidelity, implementation outcomes and audit
- 10. Adaptation control and version management
- 11. Outstanding validation and formal closure plan

Package contents at Version 1.0

Package element	Included in v1.0	Status
Initial programme theory	Context-mechanism-action-outcome model	Included
Manual and SOPs	Seven components and an integrated workflow	Included
Algorithms	Identification, review, family meeting, planning and reassessment	Included
Templates	Screening, consensus, family meeting, CACP, bedside plan, audit and adaptation log	Included
Educational resources	Orientation curriculum and competency checklist	Included

Package element	Included in v1.0	Status
Implementation workflows	Daily, weekly and event-triggered processes	Included
Formal consensus/usability evidence	Delphi, NGT trail, TIDieR and measurable usability results	Pending

1. Scope, target population and principles

1.1 Intended population

Adults in an ICU who have advanced or terminal illness, substantial risk of death or prolonged dependence, high symptom burden, uncertain or changing treatment goals, repeated critical-care use, or a need for structured communication and anticipatory planning. MCACP is needs-based and is not restricted to cancer, mechanically ventilated patients, or the final days of life.

1.2 Intended users

Intensivists, primary treating teams, palliative-care clinicians, nurses, social workers, psychologists, rehabilitation/allied-health professionals, hospital administrators, and research or quality-improvement personnel.

1.3 Governing principles

- Appropriate care is patient- and family-centred, clinically proportionate, ethically and legally defensible, and responsive to the local context [7-11].
- Prognostic scores support recognition and discussion; they must never be used as automatic futility thresholds.
- Treatment limitation is not withdrawal of care. Comfort, dignity, symptom relief, communication and family support remain active duties.
- Whenever capacity permits, the patient participates directly. When capacity is absent, the team identifies the lawful/recognised surrogate and reconstructs the patient's values and prior preferences.
- Decisions are revisited as illness, prognosis, preferences or treatment response changes.
- Core intervention functions remain stable; delivery form may be adapted to open, closed and semi-open ICU models.
- All decisions, rationale, participants, disagreements and follow-up actions are documented.

2. Development basis and initial programme theory

The package was developed using the UK Medical Research Council framework for complex interventions, realist methodology, CFIR, Normalisation Process Theory, human-centred design and structured consensus approaches [1-3,12]. Six realist reviews included 466 studies across prognostication, communication, shared decision-making, physician consensus, care delivery and care planning. Multidisciplinary stakeholder engagement included clinicians, nurses, allied health professionals, administrators, family caregivers, and technical/site experts across different ICU contexts.

Initial programme theory: When early needs-based identification is combined with serial multidisciplinary prognostication, team alignment, structured family communication, shared decision-making, comprehensive anticipatory planning, active comfort-focused care and continuous evaluation, clinicians and families develop a shared understanding of illness and available choices. This supports goal-concordant, consistent and dignified care while reducing avoidable conflict, delay and non-beneficial intervention.

Context	Mechanism activated	Expected proximal outcome	Expected longer-term outcome
Prognostic uncertainty and variable illness trajectories	Structured review and serial reassessment	Earlier recognition and shared clinical understanding	Timelier appropriate-care planning
Fragmented teams and inconsistent recommendations	Multidisciplinary alignment and role clarity	Consistent message and reduced professional uncertainty	More defensible, coordinated decisions

Context	Mechanism activated	Expected proximal outcome	Expected longer-term outcome
Family distress, uncertainty and information mismatch	Repeated structured communication and values exploration	Trust, realistic understanding and lower decisional conflict	Greater goal concordance and family satisfaction
Crisis-driven care and poor documentation	Anticipatory planning and explicit care ceilings	Continuity across shifts and teams	Fewer unwanted or non-beneficial interventions
Variable comfort care and weak feedback systems	Protocols, bedside plans, fidelity monitoring and feedback	Reliable delivery and visible gaps	Improved comfort, quality of dying and sustainability

3. Entry criteria and identification workflow

3.1 Trigger domains

Domain	Illustrative trigger - prompts review, not an automatic decision
Advanced illness	Progressive incurable illness, frailty or advanced organ disease with limited reversibility
Acute severity	High or escalating organ support, refractory deterioration, or clinician concern about survival
Trajectory	Repeated ICU admissions, prolonged ICU stay, failure to improve as expected, or recurrent treatment escalation
Communication/ decision need	Uncertain goals, disagreement, unrealistic expectations, absent prior preferences, or repeated family questions
Symptom/support need	Uncontrolled pain, dyspnoea, agitation, delirium, psychosocial or spiritual distress

Domain	Illustrative trigger - prompts review, not an automatic decision
Clinician prompt	The surprise question: 'Would I be surprised if this patient died during this admission or in the coming months?' - a 'No' response prompts review

3.2 Identification SOP

6. Screen all new admissions within 24 hours and repeat after major deterioration, treatment failure, ICU day 3-5, or a new decision point.
7. If one or more triggers are present, notify the ICU clinical lead and document the reason for review.
8. Complete a structured clinical and prognostic review using diagnosis, trajectory, organ support, reversibility, function/frailty, comorbidities and patient values.
9. Convene multidisciplinary alignment before communicating major recommendations to the family.
10. Initiate or repeat the family meeting and comprehensive anticipatory care plan.
11. Review the plan daily and after any clinically meaningful change.

Safety rule: No individual trigger, prediction score, age threshold, diagnosis or cost consideration determines treatment limitation. Decisions require clinical judgement, proportionality, patient values, multidisciplinary review and documented communication.

4. The seven MCACP components

The component specifications draw on family-centred ICU guidance, shared decision-making policy, contemporary Indian end-of-life and palliative care consensus, and the legal framework governing advance directives and the foregoing of life-sustaining treatment [7-11].

4.1 Prognostication

Core function: Create a shared, revisable understanding of likely outcomes and uncertainty.

Minimum actions

- Needs-based screening
- Structured assessment of reversibility, trajectory, function/frailty and organ support
- Use scores only as supplementary evidence
- Document best-case, worst-case and most-likely scenarios
- Set a defined reassessment time

Associated tools

Trigger screening form; Prognostic review note; Time-limited treatment-trial plan.

Minimum fidelity indicators

Eligible patients screened; Prognosis reviewed and documented; Reassessment date recorded.

4.2 Physician consensus and multidisciplinary alignment

Core function: Align the clinical team before major communication or treatment decisions.

Minimum actions

- Identify the accountable ICU lead and relevant primary/specialty teams
- Invite nursing and palliative-care input
- Record areas of agreement, uncertainty and disagreement
- Escalate unresolved conflict through a second opinion, medical board or ethics/legal route
- Nominate one communication lead

Associated tools

Multidisciplinary consensus note; Disagreement escalation pathway.

Minimum fidelity indicators

Team review completed; Single documented recommendation/message; Unresolved disagreement escalated.

4.3 Communication

Core function: Provide timely, honest, compassionate and consistent information.

Minimum actions

- Hold an initial family meeting when triggers are identified
- Assess what the patient/family understands and wants to know
- Explain uncertainty using plain language and scenarios
- Use teach-back and allow repeated discussions
- Identify family spokesperson and interpreter needs
- Document attendees, information shared, concerns and follow-up

Associated tools

Family-meeting guide; Family-meeting record; Communication prompt card.

Minimum fidelity indicators

Family meeting documented; Understanding checked; Follow-up date/owner recorded.

4.4 Shared decision-making

Core function: Reach decisions that integrate evidence, proportionality and patient values.

Minimum actions

- Assess patient decision-making capacity
- Elicit prior wishes, values, acceptable outcomes and trade-offs
- Clarify the surrogate when the patient lacks capacity
- Present reasonable options, including comfort-focused care
- Make a recommendation when families want clinical guidance
- Document the decision, rationale and review conditions

Associated tools

Shared decision-making checklist; Values and preferences prompt.

Minimum fidelity indicators

Capacity/surrogate documented; Values elicited; Decision and rationale recorded.

4.5 Comprehensive anticipatory care planning

Core function: Translate goals into specific, visible and reviewable clinical orders.

Minimum actions

- State overall goal of care
- Document treatment ceilings and escalation/non-escalation decisions
- Specify cardiopulmonary resuscitation status and organ-support plans
- Plan symptom relief, nutrition/hydration, investigations and monitoring
- Plan ward/home/hospice/palliative-care transitions where appropriate
- Set review triggers, date and responsible clinician

Associated tools

CACP form; Bedside action plan; Transfer/handover checklist.

Minimum fidelity indicators

CACP completed; Treatment ceilings visible; Plan handed over across shifts.

4.6 Care delivery

Core function: Deliver active, coordinated care focused on comfort, dignity and agreed goals.

Minimum actions

- Assess and treat pain, dyspnoea, secretions, agitation, anxiety, delirium and other distress
- De-escalate burdensome monitoring/interventions when inconsistent with goals
- Maintain nursing comfort measures, privacy and family access where feasible
- Provide psychosocial, spiritual and bereavement support
- Involve specialist palliative care according to need and availability
- Continue active care after treatment limitation

Associated tools

Symptom and comfort checklist; Nursing bedside plan; Palliative-care referral pathway; After-death care checklist.

Minimum fidelity indicators

Symptoms assessed and managed; Palliative-care need addressed; Family support documented.

4.7 Care evaluation

Core function: Measure delivery, experience, outcomes and resource use to support improvement.

Minimum actions

- Track reach, adoption, fidelity and timeliness
- Use AIM, IAM and FIM for implementation outcomes
- Measure family satisfaction and quality of dying where applicable
- Monitor ICU/hospital use, organ support, costs and DAMA
- Review data with teams and agree corrective actions
- Interpret preliminary trends cautiously and verify missing or zero values

Associated tools

Fidelity checklist; Monthly dashboard; Audit-and-feedback record.

Minimum fidelity indicators

Monthly review completed; Missing data reconciled; Improvement action assigned.

5. Integrated clinical workflow

When	Required action	Primary owner	Required record
Within 24 h of ICU admission	Screen for MCACP triggers and support needs	Research/ clinical nurse with ICU clinician	Screening form
On trigger identification	Structured prognostic assessment and team notification	ICU lead	Prognostic review
Before major family discussion	Multidisciplinary alignment; resolve or record differences	ICU lead + treating teams	Consensus note
At first decision point	Structured family meeting and values exploration	Communication lead	Family-meeting record
After agreement	Complete CACP and bedside action plan	ICU lead + nurse	CACP and orders
Every shift/day	Deliver comfort care; check plan visibility and new concerns	Bedside team	Daily plan/checklist
After deterioration or improvement	Repeat prognosis, communication and plan	ICU lead	Updated dated plan
At transfer/discharge/death	Handover, continuity and family/after-death support	Clinical team	Transition/after-death checklist
Monthly	Fidelity and outcome audit with feedback	Site implementation lead	Dashboard/action log

6. Team roles and governance

Role	Minimum responsibility
ICU clinical lead	Clinical accountability, prognostic review, team alignment, recommendation and escalation
Primary/specialty team	Disease-specific assessment, reversibility and continuity of the agreed plan
Palliative-care team	Complex symptom control, communication support, psychosocial care and transition planning
Nursing team	Trigger recognition, symptom assessment, bedside knowledge, plan implementation and continuity
Social worker/psychologist	Family assessment, practical/psychological support and follow-up
Site implementation lead	Workflow integration, training, fidelity review, adaptation control and issue resolution
Research/evaluation team	Consent/data processes, quality assurance, implementation outcomes and mixed-methods evaluation
Hospital governance	Policy, legal/ethics escalation, staffing support and audit oversight

Escalation pathway for disagreement

12. Clarify whether disagreement concerns facts, prognosis, values, roles or legal interpretation.
13. Repeat multidisciplinary case review with the relevant teams and nursing input.
14. Seek palliative-care consultation and an independent senior clinical opinion.
15. Hold a further family meeting with consistent representation and documented uncertainty.
16. If unresolved, activate the institutional medical board, ethics and/or legal-support pathway.
17. Maintain non-abandonment, symptom control and time-limited review while resolution is pursued.

7. Implementation tools and documentation templates

Tool 1. MCACP screening and activation form

- ☐ Advanced, progressive or terminal illness
- ☐ High mortality risk or escalating organ support
- ☐ Failure to improve / prolonged or repeated ICU use
- ☐ High symptom or psychosocial burden
- ☐ Uncertain goals, conflict or major decision pending
- ☐ Clinician would not be surprised by death during this admission or coming months

Field	Record
Patient identifier / ICU / date	
Trigger(s) and supporting evidence	
Immediate symptom or support needs	
ICU lead notified - name/time	
Prognostic review due by	
Family contact/spokesperson/interpreter need	
Screened by/signature	

Tool 2. Structured prognostic review

- Diagnosis, acute problem and response to current treatment
- Baseline function, frailty and comorbidity
- Current organ support and likely reversibility
- Relevant objective scores - interpreted with clinical judgement
- Best-case, most-likely and worst-case scenarios
- Key uncertainties and information still needed

- Time-limited trial: treatment, duration and success/failure criteria
- Date/time for reassessment

Tool 3. Multidisciplinary consensus note

Field	Record
Participants and roles	
Shared assessment of prognosis/reversibility	
Areas of agreement	
Areas of uncertainty/disagreement	
Recommended goal and treatment plan	
Communication lead	
Escalation required	Yes / No; route:
Review date/time	

Tool 4. Family-meeting guide

18. Prepare: align the team; identify participants, spokesperson, interpreter and setting.
19. Connect: introduce roles; acknowledge emotion and immediate concerns.
20. Assess understanding: ask what the family understands and how much detail they want.
21. Share: explain illness, response and uncertainty in plain language using scenarios.
22. Explore: ask what matters to the patient, prior wishes, acceptable outcomes and fears.
23. Deliberate: discuss reasonable options, burdens and benefits; provide a recommendation where appropriate.
24. Agree: summarise decisions, unresolved questions and the next review point.
25. Close: use teach-back, provide contact details and document the meeting.

Tool 5. Family-meeting record

Field	Record
Date/time/location	
Patient capacity and participation	
Family/surrogate participants	
Clinical participants	
Current understanding	
Information and uncertainty discussed	
Patient values/preferences	
Options, burdens and benefits discussed	
Decision/recommendation	
Emotional, spiritual or practical needs	
Teach-back / questions	
Next action, owner and review date	

Tool 6. Shared decision-making checklist

- ☐ Decision-making capacity assessed
- ☐ Patient involved to the extent possible
- ☐ Recognised surrogate/family representative identified
- ☐ Prior expressed wishes and advance directive sought
- ☐ Values, acceptable outcomes and trade-offs elicited
- ☐ Reasonable options and uncertainty explained
- ☐ Clinical recommendation offered where appropriate

☐ Agreement or disagreement documented

☐ Decision review conditions recorded

Tool 7. Comprehensive anticipatory care plan

Planning domain	Agreed plan
Overall goal of care	Recovery-focused / time-limited trial / comfort-focused / mixed
CPR status	
Intubation/ventilation	
Vasopressors and other organ support	
Dialysis / procedures / surgery	
Investigations and monitoring	
Antimicrobials, nutrition and hydration	
Symptom-management priorities	
Palliative-care involvement	
Psychosocial/spiritual/family support	
Preferred transition/disposition	
Review trigger/date/responsible clinician	

Tool 8. Daily bedside action plan

☐ Goal of care and ceilings visible to the team

- ☐ Pain assessed and managed
- ☐ Dyspnoea/respiratory distress assessed and managed
- ☐ Agitation, anxiety and delirium assessed and managed
- ☐ Secretions/nausea/other symptoms assessed and managed
- ☐ Non-beneficial tests, monitoring or interventions reviewed
- ☐ Nursing comfort, privacy, mouth/skin care and positioning addressed
- ☐ Family communication update completed or scheduled
- ☐ Palliative/psychosocial/spiritual support needs addressed
- ☐ Plan reviewed after clinical change

Clinical safety: Medication selection, dosing, routes and monitoring must follow the treating clinician's judgement, institutional policy and patient-specific factors. This package does not replace local prescribing guidance.

Tool 9. Transition and after-death care checklist

- ☐ Goals and ceilings communicated at transfer
- ☐ Medication and symptom plan handed over
- ☐ Receiving team/service confirmed
- ☐ Family informed about what to expect and whom to contact
- ☐ Equipment, home-care, hospice or palliative-care needs coordinated
- ☐ After death: confirmation, respectful care and institutional procedures completed
- ☐ Family provided practical information and bereavement support/referral
- ☐ Team debrief offered after a difficult case

8. Training and competency package

8.1 Core orientation curriculum

Module	Content	Suggested method
1. Appropriate care in ICU	Principles, ethics, legal context, non-abandonment	Brief interactive session
2. Identification and prognostication	Triggers, uncertainty, scenarios, time-limited trials	Case-based workshop

Module	Content	Suggested method
3. Team consensus	Role clarity, nursing input, disagreement management	Simulation/case huddle
4. Communication	Serious-news communication, emotion, teach-back, family meetings	Role-play
5. Shared decision-making	Capacity, surrogate decisions, values and recommendations	Case discussion
6. CACP and care delivery	Treatment ceilings, comfort care, transitions and after-death support	Bedside demonstration
7. Documentation and evaluation	Forms, fidelity, AIM/IAM/FIM, audit and adaptation log	Practical exercise

8.2 Minimum competency confirmation

- ☐ Can identify an MCACP trigger without using it as an automatic futility rule
- ☐ Can present prognostic uncertainty using best/most-likely/worst-case scenarios
- ☐ Can contribute to or lead a multidisciplinary alignment discussion
- ☐ Can conduct and document a structured family meeting
- ☐ Can elicit values and support shared decision-making
- ☐ Can complete and hand over a CACP
- ☐ Can distinguish adaptation from protocol deviation
- ☐ Can complete fidelity documentation and respond to audit feedback

9. Fidelity, implementation outcomes and audit

The evaluation domains follow established implementation-outcome taxonomy and validated brief measures of acceptability, appropriateness and feasibility [5,6].

9.1 Fidelity domains

Domain	Minimum measure	Suggested source
Reach	Eligible patients screened / eligible admissions	Screening register

Domain	Minimum measure	Suggested source
Activation	Triggered patients entering MCACP workflow	Activation log
Prognostication	Structured review and reassessment documented	Clinical record
Consensus	Multidisciplinary review before major decision	Consensus note
Communication	Family meeting, understanding and follow-up documented	Meeting record
Shared decision-making	Capacity/surrogate, values and decision documented	SDM checklist
Planning	CACP and treatment ceilings completed and visible	CACP/clinical orders
Care delivery	Symptoms and family-support needs assessed/actioned	Daily checklist
Evaluation	Monthly audit completed and corrective action assigned	Dashboard/action log

9.2 Implementation outcomes

Assess acceptability (AIM), appropriateness (IAM), feasibility (FIM), adoption, fidelity, usability, workflow integration, staff engagement and training completion. Outcome interpretation must specify numerator/denominator and distinguish true zero activity from missing or incomplete data.

9.3 Clinical, family and resource outcomes

- Symptom burden and comfort
- Quality of dying and death
- Family satisfaction and decision experience
- Palliative-care referral/consultation and communication indicators
- Organ support, ICU/hospital utilisation and length of stay
- Care costs and resource use

- Discharge against medical advice
- Mortality and disposition

Interpretation rule: Pilot trends are descriptive and hypothesis-generating. They must not be presented as causal evidence that MCACP reduces mortality, cost, resource use or DAMA without an appropriate comparative analysis.

10. Adaptation control and version management

10.1 Core function versus adaptable form

Core function - preserve	Adaptable form - may vary with documentation
Early needs-based identification	Who screens; paper or electronic prompt; timing within local rounds
Multidisciplinary alignment	Bedside huddle, formal meeting, telephone or virtual review
Structured repeated communication	Meeting timing, personnel, language and locally suitable guide
Values-based shared decision-making	Specific prompt wording and decision-aid format
Explicit anticipatory plan and treatment ceilings	Local form, order set and location in the record
Active comfort-focused care	Local symptom protocols and specialist-referral model
Evaluation and feedback	Dashboard format, meeting frequency and responsible team

10.2 Adaptation log

Date/ site	Component	Change and evidence source	Rationale/ expected effect	Core function or form?	Approval/ version

10.3 Adaptation versus protocol deviation

An adaptation changes the operational form while preserving the sanctioned objective, core function, study design and timeline; it is planned, justified and logged. A protocol deviation departs from an approved requirement or procedure and must be documented, assessed for impact and reported through the applicable governance pathway.

11. Outstanding validation and formal closure plan

Formal closure will use complete intervention reporting and transparent documentation of consensus methods [4,13].

Requirement	Evidence required for closure	Status at 31 Aug 2026
Freeze initial package	Dated version, contents register and approval record	This v1.0 document
Two-step Delphi	Panel composition, rounds, response rates, thresholds and item decisions	Pending
NGT-informed expert process	Separate meeting records, participants, outputs and decision trail	Pending formal consolidation
TIDieR description	Materials, procedures, providers, mode, setting, dose, tailoring, modifications and fidelity	Pending finalisation

Requirement	Evidence required for closure	Status at 31 Aug 2026
Frontline usability testing	Participants, task completion, time, item-level feedback and measurable usability/acceptability/appropriateness/feasibility	Pending
Revision and approval	Change log, core/adaptable classification and steering approval	Next controlled version

Proposed acceptance criteria for Version 1.1

- Delphi item-retention and consensus thresholds are prospectively specified and reported.
- All seven components have approved core functions, minimum actions and fidelity indicators.
- All forms are tested by representative frontline users from both pilot ICU models.
- Critical tasks can be completed without unresolvable workflow barriers; completion time and error points are recorded.
- AIM, IAM and FIM results and qualitative feedback are triangulated and linked to documented revisions.
- The TIDieR checklist and adaptation log match the released package version.
- The programme steering group approves the version for continued pilot implementation.

References

1. Skivington K, Matthews L, Simpson SA, et al. A new framework for developing and evaluating complex interventions: update of Medical Research Council guidance. *BMJ*. 2021;374:n2061. doi:10.1136/bmj.n2061.
2. Damschroder LJ, Aron DC, Keith RE, Kirsh SR, Alexander JA, Lowery JC. Fostering implementation of health services research findings into practice: a consolidated framework for advancing implementation science. *Implementation Science*. 2009;4:50. doi:10.1186/1748-5908-4-50.
3. May C, Finch T, Mair F, et al. Development of a theory of implementation and integration: Normalization Process Theory. *Implementation Science*. 2009;4:29. doi:10.1186/1748-5908-4-29.
4. Hoffmann TC, Glasziou PP, Boutron I, et al. Better reporting of interventions: Template for Intervention Description and Replication (TIDieR) checklist and guide. *BMJ*. 2014;348:g1687. doi:10.1136/bmj.g1687.
5. Proctor E, Silmere H, Raghavan R, et al. Outcomes for implementation research: conceptual distinctions, measurement challenges, and research agenda. *Administration and Policy in Mental Health*. 2011;38(2):65-76. doi:10.1007/s10488-010-0319-7.
6. Weiner BJ, Lewis CC, Stanick C, et al. Psychometric assessment of three newly developed implementation outcome measures. *Implementation Science*. 2017;12:108. doi:10.1186/s13012-017-0635-3.
7. Davidson JE, Aslakson RA, Long AC, et al. Guidelines for family-centered care in the neonatal, pediatric, and adult ICU. *Critical Care Medicine*. 2017;45(1):103-128. doi:10.1097/CCM.0000000000002169.
8. Kon AA, Davidson JE, Morrison W, Danis M, White DB. Shared decision making in ICUs: an American College of Critical Care Medicine and American Thoracic Society policy statement. *Critical Care Medicine*. 2016;44(1):188-201. doi:10.1097/CCM.0000000000001396.
9. Mani RK, Bhatnagar S, Butola S, et al. Indian Society of Critical Care Medicine and Indian Association of Palliative Care expert consensus and position statements for end-of-life and palliative care in the intensive care unit. *Indian Journal of Critical Care Medicine*. 2024;28(3):200-250. doi:10.5005/jp-journals-10071-24661.
10. Supreme Court of India. Common Cause (A Registered Society) v Union of India and Another. Writ Petition (Civil) No. 215 of 2005. Judgment dated 9 March 2018.
11. Supreme Court of India. Common Cause (A Registered Society) v Union of India and Another. Miscellaneous Application No. 1699 of 2019 in Writ Petition (Civil) No. 215 of 2005. Order dated 24 January 2023.

12. Pawson R, Greenhalgh T, Harvey G, Walshe K. Realist review - a new method of systematic review designed for complex policy interventions. *Journal of Health Services Research & Policy*. 2005;10 Suppl 1:21-34. doi:10.1258/1355819054308530.

13. Jünger S, Payne SA, Brine J, Radbruch L, Brearley SG. Guidance on Conducting and REporting DElphi Studies (CREDES) in palliative care: recommendations based on a methodological systematic review. *Palliative Medicine*. 2017;31(8):684-706. doi:10.1177/0269216317690685.

Approval and release record

Role	Name	Decision/ signature	Date
Principal Investigator	Dr Naveen Salins		
Co-Principal Investigator / programme representative			
KMC site implementation lead			
SJMCH site implementation lead			
Methodology/ implementation-science representative			
Document controller			

End of controlled document.

Annexure 6. Case Record Proforma

Project Title: Implementing a Multifaceted, Comprehensive, Appropriate Care Package for People with Advanced and Terminal Illnesses Needing ICU Support and Evaluating Its Benefits for Quality of Dying, Family Satisfaction, Care Cost and Resource Utilisation

Site Name: _____

Date of Entry: __/__/____

Case ID: _____

SECTION A: Patient Identification and Demographics

Hospital ID: _____

Name (Initials only): _____

Age: _____

Gender: ☐ Male ☐ Female ☐ Other

Socioeconomic Status: ☐ Below Poverty Line ☐ Middle Income ☐ High Income

Payment Source for ICU Care: ☐ Out-of-Pocket ☐ Public Scheme ☐ Insurance ☐ Other: _____

SECTION B: ICU Admission Details

Date of ICU Admission: __/__/____

Admitted from: ☐ Ward ☐ Emergency ☐ OT ☐ Other: _____

Primary Diagnosis: _____

Comorbidities: _____

Illness Trajectory Type: ☐ Acute illness ☐ Cancer-related ☐ Chronic progressive organ impairment ☐

Frailty/neurodegenerative

SECTION C: Prognostication & Eligibility

ENDING-S Score: _____

EOL Phase Identified: ☐ Yes ☐ No

Date of Scoring: __/__/____

Prognosis Documented in Notes: ☐ Yes ☐ No

Triggers for Palliative Care Referral: ☐ Prolonged ICU stay ☐ Sepsis ☐ Mechanical Ventilation ☐

Vasopressors ☐ Other: _____

SECTION D: Communication & Family Engagement

Family Meeting Conducted: ☐ Yes ☐ No

Date of First Family Meeting: ____/____/____

Who Participated: ☐ ICU Physician ☐ Palliative Care Physician ☐ Nurse ☐ Social Worker ☐ Family

Goals of Care Discussed: ☐ Yes ☐ No

Advance Directives Present: ☐ Yes ☐ No

Bereavement Support Informed: ☐ Yes ☐ No

SECTION E: Treatment Limitation Decisions

Limitation of Treatment Applied: ☐ DNAR ☐ Withholding ☐ Withdrawal ☐ Non-Escalation ☐ De-escalation

Date of Decision: ____/____/____

Documentation Done: ☐ Yes ☐ No

Critical Care Review Board (CCRB) Involved: ☐ Yes ☐ No

SECTION F: Symptom Management & End-of-Life Care

Opioid Administered: ☐ Yes ☐ No – Type: ____

Sedatives Used: ☐ Yes ☐ No – Type: ____

Symptom Checklist Used: ☐ Yes ☐ No

Compassionate Extubation Done: ☐ Yes ☐ No

Terminal Sedation Done: ☐ Yes ☐ No

After-Death Care Preferences Elicited: ☐ Yes ☐ No

Cultural/Religious Preferences Respected: ☐ Yes ☐ No

SECTION G: Outcome & Discharge Details

ICU Outcome: ☐ Died in ICU ☐ Shifted to Ward ☐ DAMA ☐ Discharged to Home ☐ Hospice Referral

Length of ICU Stay: ____ days

Duration from PC Referral to Death/Discharge: ____ days

Documentation of QODD (ICU-QODD): ☐ Done ☐ Not Done

FS-ICU Survey Collected (3–4 weeks later): ☐ Yes ☐ No

SECTION H: Additional Notes

Observations:

Name of ICU Nurse/Project Nurse completing the form: _____

Signature: _____

Date: __/__/____

Quality of Dying and Death (QODD) — is an instrument for assessing the quality of dying and death. This version is designed for self-administration by a significant other (family member and/or friend) of a patient who died in a hospital or ICU setting.

Please use this link [Microsoft Word - familyqodd 3.2.doc](#)

Adapted from -

*Glavan BJ, Engelberg RA, Downey L, Curtis JR. Using the medical record to evaluate the quality of end-of- life care in the intensive care unit. Crit Care Med. 2008 Apr;36(4):1138-46.

*Curtis JR, Downey L, Engelberg RA. The quality of dying and death: is it ready for use as an outcome measure? Chest. 2013 Feb 1;143(2):289-91. doi: 10.1378/chest.12-1941.

Family Satisfaction with Care in the Intensive Care Unit: FS-ICU 24R©

How are we doing?

We would like to hear your opinions about your family member’s recent admission to the Intensive Care Unit (ICU)

Your family member was a patient in this ICU. The questions that follow ask **YOU** about your family member’s **most recent ICU admission**. We understand that there were probably many doctors and nurses and other staff involved in caring for your family member. We know that there may be exceptions but we are interested in **your overall assessment** of the quality of care we delivered. We understand that this was probably a very difficult time for you and your family members. We would appreciate you taking the time to provide us with your opinion. Please take a moment to tell us what we did well and what we can do to make our ICU better. **Please be assured that all responses are confidential.** The Doctors and Nurses who looked after your family member will not be able to identify

PART 1: SATISFACTION WITH CARE

Please check one box that best reflects your views. If the question does not apply to your family member’s stay then check the ‘Not Applicable’ box (N/A).

HOW DID WE TREAT YOUR FAMILY MEMBER (THE PATIENT)? HOW SATISFIED ARE YOU WITH...

1.Concern and Caring by ICU Staff?

The courtesy, respect and compassion your family member (the patient) was given.

2.Sym

me

Very

Slightl

Mostly

Ver

pt

nt?

Dissati

y

Satisfie d

y

om

sfie d

Dissati

Sati

Ma

sfie d

sfie

na

d

ge

113

Completely Satisfied

☐5

N/A

☐

How well the ICU staff assessed and treated **your family member's** symptoms.

a. How well the ICU staff assessed and treated your family member's pain.

Very Dissatisfied Slightly Dissatisfied Mostly Satisfied Very Satisfied Completely Satisfied N/A

b. How well the ICU staff assessed and treated your family member's breathlessness.

Very Dissatisfied Slightly Dissatisfied Mostly Satisfied Very Satisfied Completely Satisfied N/A

☐5

N/A

☐



c.	How well the ICU staff assessed and treated your family member's agitation.	Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
		☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐

HOW DID WE TREAT YOU? HOW SATISFIED ARE YOU WITH...

3.	Consideration of your needs? How well the ICU staff showed an interest in your needs.						
		Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
		☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐
4.	Emotional support? How well the ICU staff provided emotional support to you .						
		Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
		☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐
5.	Co-ordination of care? The teamwork of all the ICU staff that took care of your family member.						
		Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
		☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐
6.	Concern and Caring by ICU Staff? The courtesy, respect and compassion you were given.						
		Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
		☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐

THE ICU STAFF: HOW SATISFIED ARE YOU WITH...

7.	Skill and Competence of ICU Nurses? How well the nurses cared for your family member.						
		Very Dissatisfied	Slightly Dissatisfied	Mostly Satisfied	Very Satisfied	Completely Satisfied	N/A
		☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	☐

8. **Frequency of Communication
With ICU Nurses?**

How often nurses communicated to
you about your family member's
condition.

d

☐1



Very

Dissatisfie

d

☐1

d

☐2



Slightly

Dissatisfie

d

☐2

d

☐3



Mostly

Satisfie

d

☐3

d

☐4



Very

Satisfie

d

☐4

☐5



Completel

y Satisfied

☐5

N/A

☐

9. **Skill and Competence of ICU Doctors (All Doctors including Residents)?**

How well doctors cared for your family member.



Very

Dissatisfied
□1



Slightly

Dissatisfied
□2



Mostly

Satisfied
□3



Very

Satisfied
□4



Completely

Satisfied
□5

N/A

□

THE WAITING ROOM

10. How satisfied are you with the atmosphere (mood) in the ICU waiting room?



Very

Dissatisfied
□1



Slightly

Dissatisfied
□2



Mostly

Satisfied
□3



Very

Satisfied
□4



Completely

Satisfied
□5

N/A

□

THE ICU

11. How satisfied are you with the atmosphere (mood) of the ICU?



Very

Dissatisfied
□1



Slightly

Dissatisfied
□2



Mostly

Satisfied
□3



Very

Satisfied
□4



Completely

Satisfied
□5

N/A

□

12. How satisfied are you with your participation in daily rounds?



Very

Dissatisfied
□1



Slightly

Dissatisfied
□2



Mostly

Satisfied
□3



Very

Satisfied
□4



Completely

Satisfied
□5

N/A

□

13. How satisfied are you with your participation in the care of your critically ill family member?



Very

Dissatisfied
□1



Slightly

Dissatisfied
□2



Mostly

Satisfied
□3



Very

Satisfied
□4



Completely

Satisfied
□5

N/A

□



14. Some people want everything done for their health problems while others do not want a lot done.



How

satisfied are you with the LEVEL or
amount of health care your family
member received in the ICU?

Very

Slightly

Mostly

Very

Completely

N/A

Dissatisfied

Dissatisfied

Satisfied

Satisfied

y Satisfied

☐1

☐2

☐3

☐4

☐5

☐

PART 2: FAMILY SATISFACTION WITH DECISION-MAKING AROUND CARE OF CRITICALLY ILL PATIENTS

INSTRUCTIONS FOR FAMILY OF CRITICALLY ILL PATIENTS

This part of the questionnaire is designed to measure how you feel about YOUR involvement in decisions related to your family member's health care. In the ICU, your family member may have received care from different people. We would like you to think about all the care your family member received when you are answering the questions.

Please check one box that best reflects your views. If the question does not apply to your family member's stay then check the 'Not Applicable' box (N/A)

INFORMATION NEEDS

HOW SATISFIED ARE YOU WITH...

15. Frequency of Communication						
With ICU Doctors?	Very	Slightly	Mostly	Very	Complete	N/A
How often doctors communicated to you about your family member's condition.	Dissatisfied	Dissatisfied	Satisfied	Satisfied	ly Satisfied	☐
	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	
16. Ease of getting information?						
Willingness of ICU staff to answer your questions.	Very	Slightly	Mostly	Very	Complete	N/A
	Dissatisfied	Dissatisfied	Satisfied	Satisfied	ly Satisfied	☐
	☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	
17. Understanding of Information?						
How well ICU staff provided you						

with explanations that you understood.

Very	Slightly	Mostly	Very	Complete	N/A
Dissatisfied	Dissatisfied	Satisfied	Satisfied	ly	☐
ed	ed	ed	ed	Satisfied	
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	

18. **Honesty of Information?**

The honesty of information provided to you about your family member's condition.

					
Very	Slightly	Mostly	Very	Complete	N/A
Dissatisfied	Dissatisfied	Satisfied	Satisfied	ly	☐
ed	ed	ed	ed	Satisfied	
☐ 1	☐ 2	☐ 3	☐ 4	☐ 5	

19. **Completeness of Information?**

				
---	---	---	---	---

How well ICU staff informed you what was happening to your family member and why things were being done.	Very Dissatisfied ☐ 1	Slightly Dissatisfied ☐ 2	Mostly Satisfied ☐ 3	Very Satisfied ☐ 4	Completely Satisfied ☐ 5	N/A ☐
--	---	---	--	--	--	---------------------------------

20. Consistency of Information?

The consistency of information provided to you about your family member's condition (Did you get a similar story from the doctor, nurse, etc.)



Very

Dissatisfied

☐1



Slightly

Dissatisfied

☐2



Mostly

Satisfied

☐3



Very

Satisfied

☐4



Completely

Satisfied

☐5

N/A

☐

PROCESS OF MAKING DECISIONS

During your family member's stay in the ICU, many important decisions were made regarding the health care she or he received. From the following questions, select only **one** answer from each of the following set of ideas that best matches your views:

21. How included or excluded did you feel in the decision making process?

- ☐ 1 I felt very excluded
- ☐ 2 I felt somewhat excluded
- ☐ 3 I felt neither included nor excluded from the decision making process
- ☐ 4 I felt somewhat included
- ☐ 5 I felt very included

22. How supported did you feel during the decision making process?

- ☐ 1 I felt totally unsupported
- ☐ 2 I felt slightly unsupported
- ☐ 3 I felt neither unsupported nor supported
- ☐ 4 I felt supported
- ☐ 5 I felt very supported

23. Did you feel you had control over the care of your family member?

- ☐ 1 I felt really out of control and that the health care system took over and dictated the care my family member received
- ☐ 2 I felt somewhat out of control and that the health care system took over and dictated the care my family member received
- ☐ 3 I felt neither in control or out of control
- ☐ 4 I felt I had some control over the care my family member received
- ☐ 5 I felt that I had good control over the care my family member received

24. When making decisions, did you have adequate time to have your concerns addressed and questions answered?

- ☐ 1 The time I had was definitely inadequate
- ☐ 2 The time I had was slightly inadequate

- ☐ 3 The time I had was adequate
- ☐ 4 The time I had was more than adequate
- ☐ 5 I had a substantial amount of time

If your family member died during the ICU stay, please answer the following questions (25-27). If your family member did not die, please skip to question 28.

25. Which of the following best describes your views:

- ☐ 1 I felt my family member's life was prolonged unnecessarily
- ☐ 2 I felt my family member's life was slightly prolonged unnecessarily
- ☐ 3 I felt my family member's life was neither prolonged nor shortened unnecessarily
- ☐ 4 I felt my family member's life was slightly shortened unnecessarily
- ☐ 5 I felt my family member's life was shortened unnecessarily

26. During the final hours of your family member's life, which of the following best describes your views:

- ☐ 1 I felt that he/she was very uncomfortable
- ☐ 2 I felt that he/she was slightly uncomfortable
- ☐ 3 I felt that he/she was mostly comfortable
- ☐ 4 I felt that he/she was very comfortable
- ☐ 5 I felt that he/she was totally comfortable

27. During the last few hours before your family member's death, which of the following best describes your views:

- ☐ 1 I felt very abandoned by the health care team
- ☐ 2 I felt abandoned by the health care team
- ☐ 3 I felt neither abandoned nor supported by the health care team
- ☐ 4 I felt supported by the health care team
- ☐ 5 I felt very supported by the health care team

28. Do you have any suggestions on how to make care provided in the ICU better?

29. Do you have any comments on things we did well?

30. Do you have any suggestions on how we could improve the ICU experience for you and your family member?

DEMOGRAPHICS:

Please complete the following to help us know a little about you and your relationship to the patient.

1. **I am:** ☐ Male ☐ Female ☐ Other

2. **I am** _____ years old

3. **I am the patient's:**

- | | | |
|-----------------------------------|----------------------------------|--|
| ☐ Wife | ☐ Husband | ☐ Partner |
| ☐ Mother | ☐ Father | ☐ Sister ☐ Brother |
| ☐ Daughter | ☐ Son | ☐ Other (Please specify): _____ |

4. **Before this most recent event, have you been involved as a family member of a patient in an ICU (Intensive Care Unit)?**

☐ Yes ☐ No

5. **Do you currently live with the patient?** ☐ Yes ☐ No

If no, then on average how often do you see the patient?

☐ More than weekly ☐ Weekly ☐ Monthly ☐ Yearly ☐ Less than once a year

6. **Where do you live?**

☐ In the city or town where the hospital is located ☐ Out of town

7. Which of the following best describes the highest level of education you have completed?

- ☐ Did not complete secondary school or high school
- ☐ Completed secondary or high school
- ☐ Had some university education or completed a community college, technical college, or post-secondary program (for example; trade, technical or vocational school)
- ☐ University degree (for example; BA, BSc, BSN)
- ☐ Graduate degree (for example; MD, DDS, DMD, DVM, OD, Master's, or PhD)

We would like to thank you very much for your participation and your opinions.



The TIDieR (Template for Intervention Description and Replication) Checklist*:

Information to include when describing an intervention and the location of the information

Item number	Item	Where located **	
		Primary paper (page or appendix number)	Other † (details)
1.	BRIEF NAME Provide the name or a phrase that describes the intervention.	_____	_____
	WHY		
2.	Describe any rationale, theory, or goal of the elements essential to the intervention.	_____	_____
	WHAT		
3.	Materials: Describe any physical or informational materials used in the intervention, including those provided to participants or used in intervention delivery or in training of intervention providers. Provide information on where the materials can be accessed (e.g. online appendix, URL).	_____	_____
4.	Procedures: Describe each of the procedures, activities, and/or processes used in the intervention, including any enabling or support activities.	_____	_____
	WHO PROVIDED		
5.	For each category of intervention provider (e.g. psychologist, nursing assistant), describe their expertise, background and any specific training given.	_____	_____
	HOW		
6.	Describe the modes of delivery (e.g. face-to-face or by some other mechanism, such as internet or telephone) of the intervention and whether it was provided individually or in a group.	_____	_____
	WHERE		
7.	Describe the type(s) of location(s) where the intervention occurred, including any necessary infrastructure or relevant features.	_____	_____

TIDieR checklist

WHEN and HOW MUCH		
8.	Describe the number of times the intervention was delivered and over what period of time including the number of sessions, their schedule, and their duration, intensity or dose.	
TAILORING		
9.	If the intervention was planned to be personalised, titrated or adapted, then describe what, why, when, and how.	
MODIFICATIONS		
10.*	If the intervention was modified during the course of the study, describe the changes (what, why, when, and how).	
HOW WELL		
11.	Planned: If intervention adherence or fidelity was assessed, describe how and by whom, and if any strategies were used to maintain or improve fidelity, describe them.	
12.*	Actual: If intervention adherence or fidelity was assessed, describe the extent to which the intervention was delivered as planned.	

** **Authors** - use N/A if an item is not applicable for the intervention being described. **Reviewers** – use ‘?’ if information about the element is not reported/not sufficiently reported.

† If the information is not provided in the primary paper, give details of where this information is available. This may include locations such as a published protocol or other published papers (provide citation details) or a website (provide the URL).

‡ If completing the TIDieR checklist for a protocol, these items are not relevant to the protocol and cannot be described until the study is complete.

* We strongly recommend using this checklist in conjunction with the TIDieR guide (see *BMJ* 2014;348:g1687) which contains an explanation and elaboration for each item.

* The focus of TIDieR is on reporting details of the intervention elements (and where relevant, comparison elements) of a study. Other elements and methodological features of studies are covered by other reporting statements and checklists and have not been duplicated as part of the TIDieR checklist. When a randomised trial is being reported, the TIDieR checklist should be used in conjunction with the CONSORT statement (see www.consort-statement.org) as an extension of Item 5 of the CONSORT 2010 Statement. When a clinical trial protocol is being reported, the TIDieR checklist should be used in conjunction with the SPIRIT statement as an extension of Item 11 of the SPIRIT 2013 Statement (see www.spirit-statement.org). For alternate study designs, TIDieR can be used in conjunction with the appropriate checklist for that study design (see www.equator-network.org).

TIDieR checklist

List of Abbreviations

Abbreviation	Full form
AIM	Acceptability of Intervention Measure
AIIMS	All India Institute of Medical Sciences
ApTCare-ICU	Appropriate Transitions in Care for Intensive Care Unit Patients
BBN	Breaking Bad News
CACP	Comprehensive Appropriate Care Plan
CAR	Centre for Advanced Research
CCM	Critical Care Medicine
CCRB	Critical Care Review Board
CFIR	Consolidated Framework for Implementation Research
CMO	Context–Mechanism–Outcome
CMC	Christian Medical College
CPR	Cardiopulmonary Resuscitation
CREDES	Conducting and REporting of DElphi Studies
CRF	Case Record Form
DAMA	Discharge Against Medical Advice
DGHS	Directorate General of Health Services
DNI	Do Not Intubate
DNAR	Do Not Attempt Resuscitation
DNR	Do Not Resuscitate
DOR	Discharge on Request
EOL	End of Life
EOLC	End-of-Life Care
EPIC	End-of-Life Care and Palliative Care in the Intensive Care Unit course
ERIC	Expert Recommendations for Implementing Change

Abbreviation	Full form
FGD	Focus Group Discussion
FIM	Feasibility of Intervention Measure
FS-ICU	Family Satisfaction in the Intensive Care Unit
GCP	Good Clinical Practice
HCD	Human-Centred Design
HFNC	High-Flow Nasal Cannula
IAM	Intervention Appropriateness Measure
IAPC	Indian Association of Palliative Care
IAPCON	Indian Association of Palliative Care Conference
ICMR	Indian Council of Medical Research
ICU	Intensive Care Unit
ICU-PC	Intensive Care Unit–Palliative Care
IEC	Institutional Ethics Committee
IQR	Interquartile Range
ISCCM	Indian Society of Critical Care Medicine
KMC	Kasturba Medical College
MATHRU	Project name for the programme implementing and evaluating the MCACP in Indian ICUs
MCACP	Multifaceted Comprehensive Appropriate Care Package
MD	Doctor of Medicine
MDT	Multidisciplinary Team
Mini-CEX	Mini Clinical Evaluation Exercise
MRC	Medical Research Council
MSW	Medical Social Worker
NABH	National Accreditation Board for Hospitals and Healthcare Providers
NGT	Nominal Group Technique

Abbreviation	Full form
NHA	National Health Authority
NHM	National Health Mission
NIMHANS	National Institute of Mental Health and Neuro Sciences
NIV	Non-Invasive Ventilation
NPT	Normalisation Process Theory
OSCE	Objective Structured Clinical Examination
PC	Palliative Care
PDSA	Plan-Do-Study-Act
PFCC	Patient- and Family-Centred Care
PhD	Doctor of Philosophy
PMB	Primary Medical Board
PPC	Paediatric Palliative Care
QODD	Quality of Dying and Death
QI	Quality Improvement
RCT	Randomised Controlled Trial
RE-AIM	Reach, Effectiveness, Adoption, Implementation and Maintenance
SDM	Shared Decision-Making
SJMC	St John's Medical College
SJMCH	St John's Medical College Hospital
SJRI	St John's Research Institute
SMB	Secondary Medical Board
SOP	Standard Operating Procedure
SVYM	Swami Vivekananda Youth Movement
TIDieR	Template for Intervention Description and Replication
VMMC	Vardhman Mahavir Medical College
WHO	World Health Organization

Abbreviation

	Full form
WHOCC	World Health Organization Collaborating Centre
WHO-SEARO	World Health Organization Regional Office for South-East Asia
WP	Work Package

Communication mnemonics used in the training curriculum

Abbreviation

	Meaning
NURSE	Name the emotion: Understand, Respect, Support and Explore
SOLER	Sit squarely, maintain an Open posture, lean towards the person, maintain Eye contact and relax
SPIKES	Setting, Perception, Invitation, Knowledge, Emotions or Empathy, and Strategy or Summary
VALUE	Value family statements, acknowledge emotions, listen, understand the patient as a person, and elicit questions